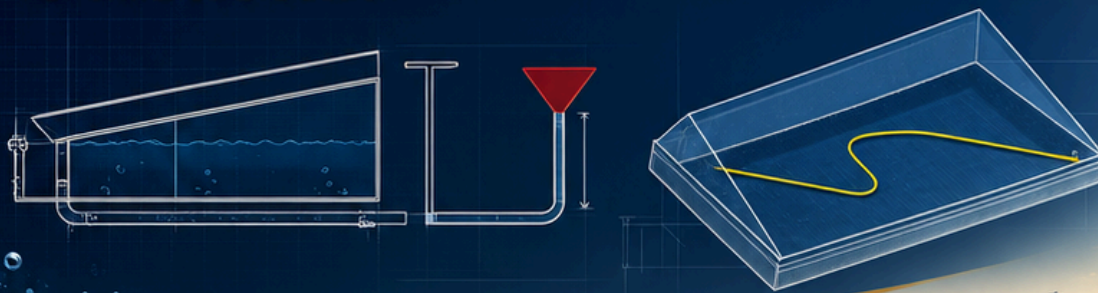


ACADEMIC MONOGRAPH

SOLAR-POWERED DESALINATION WITH FORCED-AIR BUBBLING



Design | Mathematical Modelling | Fabrication | Testing |
Performance Evaluation



ABDUL MURAD

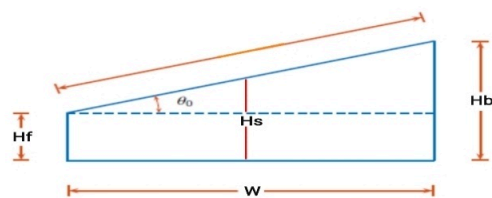
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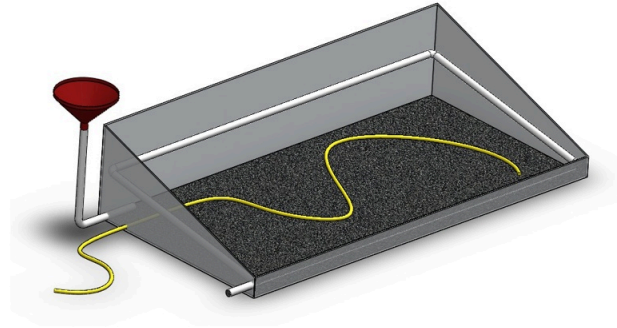
Solar-Powered Desalination with Forced-Air Bubbling

Design, Mathematical Modelling, Fabrication, Testing
and Performance Evaluation

(a)



(b)



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Preface

Purpose, provenance and academic use

This monograph presents a structured academic account of the design, mathematical modelling, fabrication, testing and performance evaluation of a solar-powered single-basin still enhanced by forced-air bubbling. The work is based on the author's undergraduate research record developed under the climatic conditions of Karachi, Pakistan.

The book has been organised for scholarly reading, research discussion and postgraduate application portfolios. Experimental data, figures, tables and numerical results are retained from the original research record, while the presentation has been reorganised into a coherent monograph format.

The underlying final-year project was undertaken in a collaborative academic environment. This single-author monograph identifies Abdul Murad as the author of this book-format adaptation; supervisory and collaborative contributions documented in the original project record remain part of that record.

Research context

Single-basin solar still | Forced-air bubbling / aeration | Energy-balance modelling | CAD and fabrication | Field testing | Productivity and economic evaluation

Suggested citation: Murad, A. (2026). Solar-Powered Desalination with Forced-Air Bubbling. Academic Monograph.

Abstract

Scope, method and principal reported findings

This monograph presents the design, fabrication and performance analysis of a single-basin solar still enhanced by forced-air bubbling. The work was developed for the climatic conditions of Karachi, Pakistan, and combines a mathematical energy-balance model with CAD development, material selection, fabrication, experimental testing and economic assessment. Comparative testing was performed with and without aeration. The reported average daily freshwater production was 2.8 L for the aerated configuration and 2.4 L for the non-aerated configuration. The corresponding reported production costs were 4.98 PKR/L and 5.38 PKR/L. The mathematical model was used to estimate the thermal behaviour of the basin, water and glass surfaces, and the predicted temperature trends were compared with experimental observations. Water-quality testing and month-by-month productivity analysis were also included in the original study. The combined findings indicate that forced-air bubbling improved the observed productivity of the tested solar still and reduced the reported unit water cost under the investigated conditions.

KEY REPORTED COMPARISON

2.8 L/day aerated | 2.4 L/day non-aerated

KEYWORDS

solar desalination | solar still | forced-air bubbling | aeration | mathematical modelling | freshwater productivity

STUDY PROFILE

Research Overview

A concise map of the experimental programme

RESEARCH SETTING

Karachi, Pakistan

SYSTEM

Single-basin solar still

ENHANCEMENT

Forced-air bubbling / aeration

DAILY FRESHWATER YIELD

2.8 L aerated | 2.4 L non-aerated

REPORTED WATER COST

4.98 PKR/L aerated | 5.38 PKR/L non-aerated

STUDY WORKFLOW

Design -> Model -> Fabricate -> Test -> Compare -> Evaluate



Experimental setup documented in the original research record.



NAVIGATION

Contents

Academic monograph structure

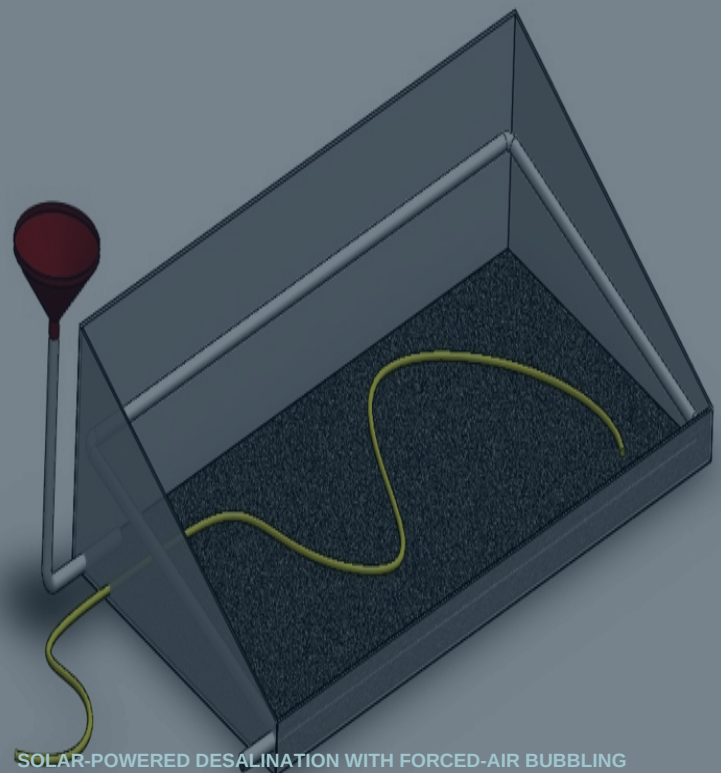
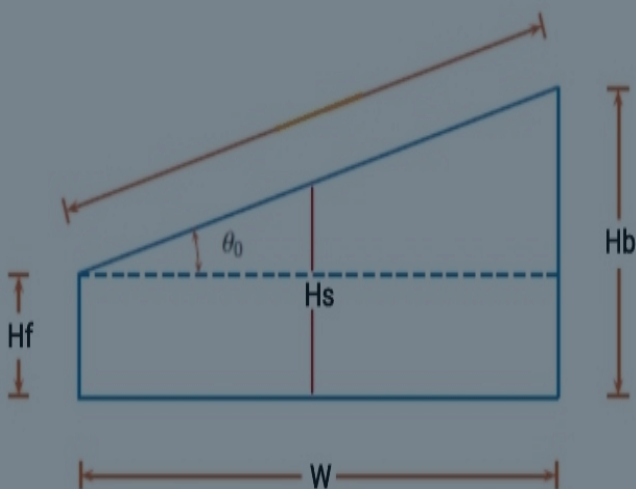
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Introduction and Foundations

Chapter 1

Water as a natural resource, core desalination concepts, conventional processes and the role of solar desalination.

(b)



CHAPTER 1.

1.1. INTRODUCTION

This chapter provides an overview of the importance of water as a natural resource, the basic concepts of desalination, various desalination techniques, and the role of solar desalination in water treatment. Shortcomings. The chapter also examines the historical development of solar imaging and highlights current research gaps in the field.

1.1.1. Considering water as a natural resource:

Water is an essential resource for all life forms and an important part of various human activities. Despite its abundance, only a small fraction of the earth's water is available and suitable for human use. About 97.5% water on earth is salty, and the rest 2.5% of fresh water is mostly contained in glaciers and ice sheets. Thus, less than 1% of the planet's water is directly available for human use. As the world's population grows and industrialization accelerates, the demand for fresh water increases at an unprecedented rate. This demand is exacerbated by climate change, which affects the distribution and availability of water resources. Therefore, there is a severe water shortage in many regions, which requires innovative solutions to increase water supply as shown in the fig 1. Mishra, Rakesh Kumar.et al [1]

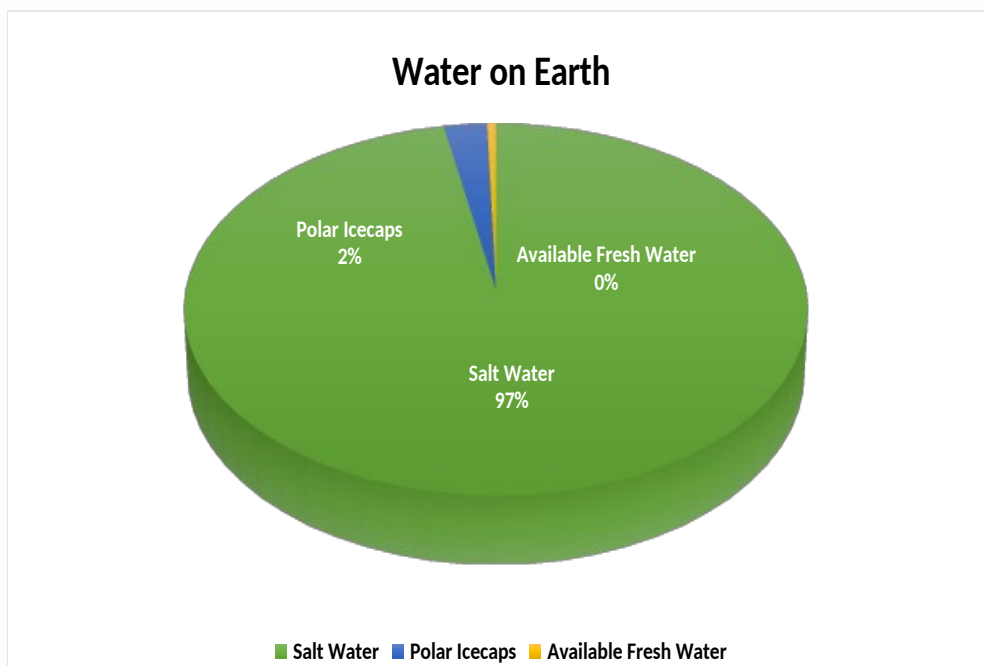


Figure 1: Shows Comparison between Fresh water and Salty Water.

1.1.2. Basic Concepts:

Desalination is a technique of removal of dissolved salts and other impurities from salt water to make fresh water suitable for human consumption, agricultural and industry use. The basic aim is the removal of salt and other minerals from water molecules using various physical and

chemical processes. Desalination techniques can be divided into two categories: heating processes and membrane processes. Almasoudi.et.al [2]

1.1.3. Overview of desalination techniques:

Numerous desalination technologies have been introduced over the times, each with their own advantages and limitations. There are the following common methods.

1.1.4. Reverse-Osmosis (RO):

Reverse osmosis (RO) is a membrane-based desalination in which salt water is forced under high pressure through a permeable membrane. The membrane pushes water molecules to pass through while blocking dissolved salts and other impurities. Harby, Ket.al. [3]. RO is widely used due to its high efficiency and ability to remove various pollutants. However, this requires significant energy input and careful control of membrane fouling.

1.1.5. Multi - Stage Flash Distillation (MSFD):

Multi-Stage Flash Distillation (MSFD) is desalination that include heating salty water to make steam and then condensed back to produce fresh water. The process is done in several steps or "flashes" to improve efficiency. MSF is highly reliable and capable of handling large desalination operations, but is energy intensive and has high operating costs. Yilmaz, Fatih, Murat Ozturk, and Resat Selbas. [4]

1.1.6. Multiple - Effect Distillation (MED):

Similar to MSFD, MED is a thermal desalination technique where salt water is heated to produce steam. The process uses multiple stages, or "effects," each operating at progressively lower pressures and temperatures to maximize energy efficiency [5]. MED is known for its lower energy consumption compared to MSF, but still requires significant thermal energy.

1.1.7. Electrodialysis (ED):

Electrodialysis is a process in which ions are transferred through ion-selective membranes with the effects of an electric field. This technology is effective for desalination of brackish water and has lower energy consumption than RO at low to moderate salinities [6]. However, ED is less effective in high salinity water and requires regular maintenance to prevent membrane fouling.

1.1.8. Solar desalination:

Solar desalination uses solar energy in the desalination process, making it an attractive option for sustainable water production. There the following types of solar desalination; direct and indirect. Direct solar desalination involves using a solar charge that mimics the natural water cycle by evaporating water and condensing the vapor into fresh water. Goosen, Mattheus [7]. Indirect solar desalination combines conventional desalination techniques with solar thermal or photovoltaic systems to produce the required energy.

Literature Review

Chapter 2

Research context for solar desalination, efficiency enhancement, materials, hybrid approaches, economics and water quality.



SOLAR-POWERED DESALINATION WITH FORCED-AIR BUBBLING

CHAPTER 2.

2.1. LITERATURE REVIEW

The purpose of the literature review is to provide a comprehensive analysis of recent advances in solar desalination systems. This review covers various techniques and innovations that have been investigated to improve the cost-effectiveness, efficiency and overall performance of these systems. The chapter is divided into several main research areas, each of which highlights important findings from recent research articles and papers.

2.1.1. Techniques to Enhance Solar Energy Efficiency:

As solar devices are simple and have low operation costs, their efficiency is comparatively modest, meeting ongoing researches to improve performance. A desirable approach involves integrating phase change materials (PCMs) to enhance energy storage and release by Ahmed, Hamza, et al. [8] demonstrated a 30% increase in daily water productivity by integrating PCMs into solar panels, capturing excess solar energy during peak sunlight hours and providing it during non-sunlight hours. More recently, a study by Xubo, H. A. O., Baolian, N. I. U. [9] found the usage of grapheme-based PCMs, achieving a remarkable 45% increase in solar energy efficiency. These findings show the critical importance of selecting suitable PCMs to enhance energy storage and release efficiency.

2.1.2. Nanotechnology Advancements in solar energy Desalination:

Nan technologies have emerged as an innovative field to optimize solar energy desalination. [10] Found the integration of Nano fluids in solar cells, showing a 40% increase in evaporation and water production. Nano fluids, contain basic fluids with suspended nanoparticles, giving exceptional thermal properties, which includes increased thermal conductivity and heat absorption. Building on this research, a recent study by Meng, Fantao, et al [11] explored the application of grapheme-based Nano composite in solar stills, showing a remarkable 55% improvement in water yield. These findings show the significant potential of nanotechnology to augment solar energy desalination efficiency.

2.1.3. Optimizing Solar Panel Design for Enhanced Efficiency:

The geometry and of system and materials used in solar stills play important role in determining their efficiency. Omara, Z.M et al. [12] examined a deep analysis of various design modifications, showing that double-layered solar stills with coatings on inside walls can effectively enhance distillate rates by enhancing the utilization of incoming solar radiation. Furthermore, their research showed that integrating inclined surfaces and multiple condensing surfaces can optimize the condensation process, leading to increased freshwater production. A recent study by Sun, Zerong, et al. [13] done upon this research, showing the integration of 3D-printed micro-channel condensers in solar stills, which showed a notable 28% increase in water yield. These findings underscore the essence of innovative design modifications solar panel efficiency.

2.1.4. Integrating Photovoltaic in PV Systems for Enhanced Desalination:

The use of photovoltaic in Photovoltaic (PV) systems can power auxiliary components like pumps, fans, and heaters, increasing overall efficiency. [14] Found a hybrid PV-Thermal (PVT) solar desalination system, including solar panels with thermal collectors. This innovation generates electricity while using waste heat for desalination, showing 25% increment in the production of water compared to standalone solar systems.

2.1.5. Hybrid Desalination systems for Optimized Performance:

Hybrid distill systems which combine multiple technologies, can increase the strengths of each method. Moh'd A, A. N., & Dawahdeh, A. I. et.al. [15] Explored a hybrid system incorporating solar stills and reverse osmosis (RO) technology, showed better water quality and higher production rates. This approach is simple and low energy consuming solar stills, using RO to purify the water. A recent study [16] developed a novel hybrid system combining multi-stage flash distillation (MSF) with solar-powered RO, demonstrating a significant 30% reduction in energy consumption and a 20% increase in water production.

2.1.6. Advancements in Materials for Enhanced Solar Energy Efficiency:

Material innovations, particularly in basin and covers, have the potential to significantly increase solar energy efficiency. Chauhan, V. K., & Shukla, S. K. [17] Explained a detailed review of different basin materials, including blackened aluminum, copper, and graphite, and investigated that graphite-coated absorbers showed superior heat resistance properties, showing in the highest thermal efficiency. Furthermore, the use of coated materials like low iron glass and antireflective coatings were shown to enhance the transfer of radiation to the absorber, enhancing efficiency. A recent study by Guo, Yifan, et al. [18] The application of nanostructured surfaces in solar absorbers, resulting a remarkable 15% increase in thermal efficiency due to improved light trapping and heat transfer.

2.1.7. Economical and Environmental Evaluations of Solar Desalination:

Accounting the economical and environmental reliability of solar still is important for their more adoption. Kumar, P. G., [19] Explained a detailed life cycle cost analysis, showing initial investments, maintenance, and operating costs, and found that solar desalination is economically reliable for home use applications, specifically in remote and arid areas. Moreover, their study showed that solar desalination has a significantly lower environmental effects compared to traditional methods, primarily due to reduced fossil fuel consumption and greenhouse emissions. A recent study by Shokri, A., & Fard, M. S. [20] found a techno-economic analysis of large-scale solar desalination plants, showed that they can achieve competitive costs with conventional desalination methods while maintaining a lower carbon footprint.

2.1.8. Enhancing Water Quality in Solar Desalination:

The quality of desalinated water is very important for its safe use. Poirier, Kristofer, et.al. [21] Explained different pre-treatment ways to enhance the quality of water distilled through solar stills, which are remineralization that recovers important minerals, and disinfection, which eliminates remaining pathogens. The study showed the necessity of pre-treatments to meet

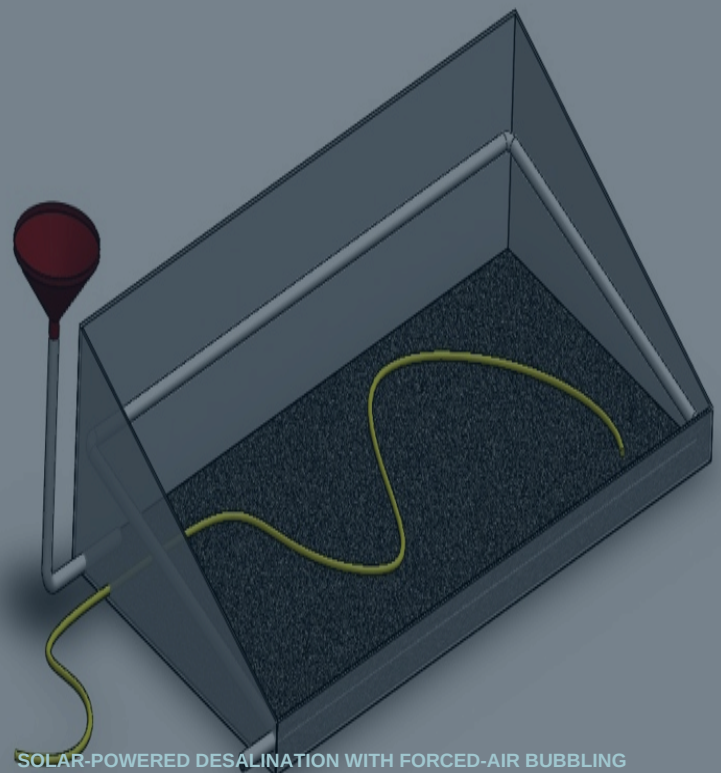
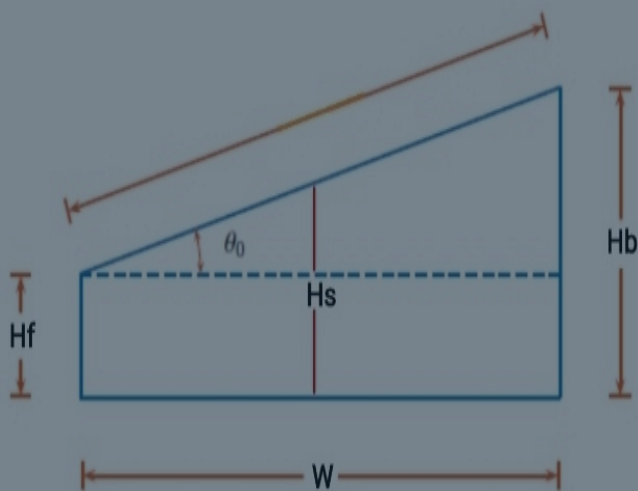
international drinkable water standards and improve the acceptability of solar desalinated water. A recent study by (Poirier, Kristofer et al, 2023 [21] showed the effectiveness of advanced oxidation processes (AOPs) as a pee-treatment method, showing significant improvements in water quality and removal of contaminants, thereby providing the production of safe and portable water.

Design Considerations & Fabrication

Chapter 3

Solar-still geometry, thermal considerations, material selection, CAD development, fabrication and the aeration arrangement.

(b)



CHAPTER 3.

3.1 DESIGN CONSIDERATIONS & FABRICATIONS

Designing a solar desalination system requires a thorough understanding of the many factors that affect its efficiency, performance and reliability. This chapter goes into the intricate details of design considerations, including solar panel geometry, materials selection, and optimization of key components to maximize freshwater production.

3.1.1. Solar Still Geometry:

The geometry of a solar still is critical to its operation and efficiency [8] it determines the surface area used for the absorption, evaporation and condensation processes of solar radiation. Several geometric parameters are important for optimal performance, including the shape, size and angle of the slope.

$$\dot{Q} = \sigma A T^4 \quad (\text{Stefan-Boltzmann Law})$$

$$\frac{\dot{Q}}{A} = -K \frac{\partial T}{\partial x} \quad (\text{Differential form of Fourier's law})$$

$$\frac{\dot{Q}}{A} = h A \Delta T \quad (\text{Natural law of coolant convective heat transfer})$$

These equations have been used for conductive, convection and radiation of heat through the solar still.

3.1.2. Thermodynamics:

At the heart of solar desalination is the thermodynamics. The solar panel must maintain favorable temperature differences to allow evaporation of salt water and subsequent condensation of fresh water. Understanding the principles of heat transfer, vapor pressure difference, and thermal equilibrium is essential to designing effective solar images. [8]

$$\Delta E = 0 \quad (\text{Closed System})$$

$$E_{in} = E_{out} \quad (\text{Open System})$$

These equations show the concepts of energy transfer into the system and energy out from the system and also help us to know about the type of system whether the system is closed or open.

3.1.3. Geometry Selection:

The selection of the geometry depends upon the reliability, functionality and market availability which has been decided after a detailed researches as follow.

a) Length (L) = 1.2m and Width = 0.6m

The selected dimensions of 1.2m length and 0.6m width of solar still designs aims to balance surface area for solar collection, construction point of view and maintenance. [8]

Front Wall Height = 0.06m and Back Wall Height = 0.27m

These wall heights are selected to provide an optimal angle for the glass cover, providing maximum solar radiation absorption and efficient water condensation on the glass surface. [8]

b) Specific Height (H)=0.165m

The selected specific height, which is the distance between the surface of water and the glass cover, maximizing the temperature difference and enhancing evaporation rates. [8] Highlighting specific height to improve to improve performance.

c) Slope Angle =19°

The angle of slope 19° is selected to ensure maximum solar incidence throughout the day, allowing condensed water to be collected in the channels. Abdelgaied, Mohamed, et al. [22]

Basin Aspect Ratio = 2

The aspect ratio of 2 (length to width) is selected to ensure efficient solar absorption and minimize heat losses and the construction practical. Ahmed, Hamza, et al. [8] emphasized the importance of the basin aspect ratio in achieving uniform temperature distribution and maximizing distillation efficiency .

1. Glass Thickness = 0.004m

A glass thickness of 4mm is selected to provide structural strength and transmissivity of solar radiation. Ahmed, Hamza, et al [8] recommended using glass with a thickness of around 4mm for balancing strength and efficiency in solar stills.

2. Ceramic Base Thickness =0.008m

A ceramic base with a thickness of 8mm is chosen for its high thermal conductivity, which helps in fast heating the water and increasing the evaporation rate. Ahmed, Hamza, et al [8] Discussed the benefits of using high thermal conductive materials for the base of the solar still to improve efficiency:

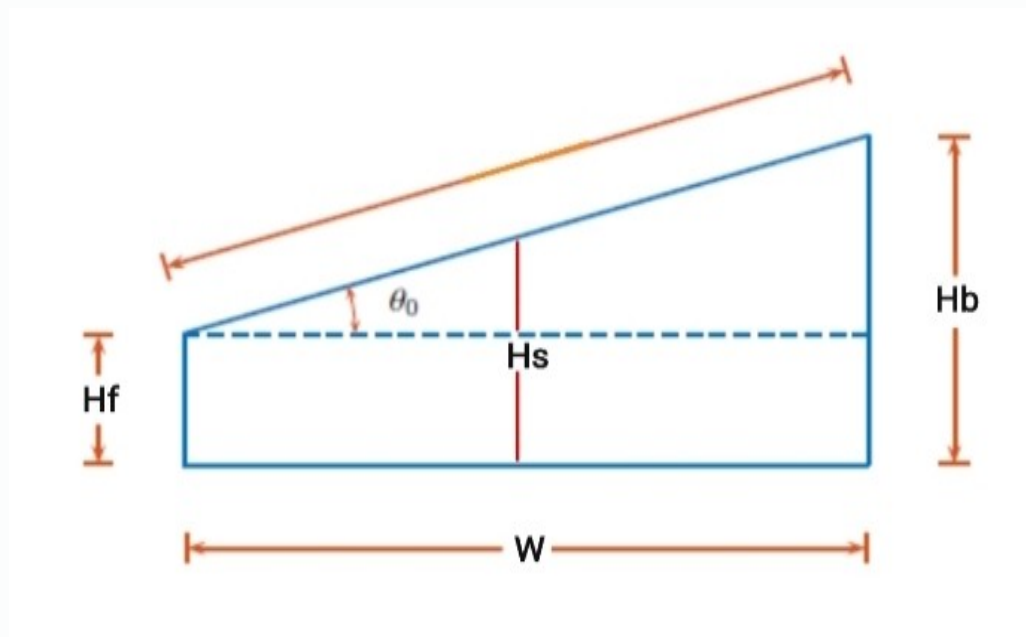


Figure 2: Shows the Geometry view from side

Table shows the dimensions of the system

Length (L)	1.2m
Width (W)	0.6m
Front Wall Height (Hf)	0.06m
Back Wall Height (Hb)	0.27m
Specific Height ($H_s = [H_f + H_b]/2$)	0.165m
Slope (Θ)	19
Basin Aspect Ratio (L/W)	2
Glass Thickness	0.004m
Ceramic base Thickness	0.008m

3.1.1. Key Considerations Include The Following Optimizations.

a) Evaporative Surface Area:

Maximizing surface area exposed to solar radiation enhance evaporation rate, which increases production of freshwater.

a) Condensing Surface:

The design of an effective condensing surface includes balanced factors such as thermal conductivity, surface roughness and wettability to provide rapid condensing of water vapor.

b) Heat loss:

Insulating the solar panel to avoid heat loss to the environment is very important to maintain a high system temperature, especially during low solar radiation.

3.1.2. Assembling the heat absorber plate:

The heat absorber plate acts as an interface between solar radiation and salt water, facilitating the conversion of solar energy into thermal energy for evaporation.

3.1.3. Plate material and geometry:

Choosing the appropriate material and geometry of the damping plate is most important to achieve effective energy absorption and heat transfer. Common materials include.

a) Blackened Ceramic Base:

Black coated ceramic plate have high solar absorption and thermal conductivity, allowing the salt to heat up quickly.

b) Transparent Covers:

Glass or plastic covers allow solar radiation to pass through and minimize heat loss, providing an efficient way to heat the damper plates.

Optimizing the geometry of the absorbent sheet includes aspects such as area, thickness and orientation to increase solar absorption and decrease heat loss.

3.1.4. Surface coatings:

Coating the surface has an important role in improving absorption capacity of the board and reducing unwanted heat losses. To enhance the performance of solar absorption, black coatings with high absorptivity and low emissivity are usually applied to the surface of the absorber. In addition, anti-reflective coatings help minimize the loss of incoming solar radiation by reducing surface reflections.

3.1.5. Absorbent material:

The choice of absorbent material affects its thermal properties, corrosion resistance and longevity in severe operating conditions.

3.1.6. Covering material:

The covering material of the solar still has several functions, including

- a. The penetration of solar radiation into the system.
- b. Minimization of heat energy loss because of convection and radiation.
- c. Facilitates the condensation of water vapor.

Common cover materials include glass, polycarbonate and acrylic, each offering unique advantages in terms of transparency, durability and thermal insulation.

3.1.7. Housing Material:

The cover material of solar still must be resistant to corrosion-, durable and heat-insulating to ensure long-term performance in various environmental conditions.

3.1.8. Introduction of aerator:

Integrating an aerator into a solar panel structure can significantly improve its performance by promoting air-water mixing and increasing the effective evaporation surface.

3.1.9. Benefits and Advantages:

Incorporating advanced design considerations, materials and components in a solar desalination system offers several benefits including:

- a) Increased freshwater production capacity.
- b) Better power efficiency.
- c) Better system reliability and longevity.
- d) Cost effective operation and maintenance.

3.2. CAD MODEL

3.2.1. Solar Collector Plate (part):

The solar collector plate, a critical component of the solar desalination system, possesses the following dimensions:

- I. Length: 120 cm
 - II. Width: 60 cm
 - III. Thickness: 0.8 cm
-

3.2.2. Glass Cover (part):

The glass cover, designed to protect the solar still while allowing sunlight to pass through, is characterized by the following dimensions:

- I. Length: 124 cm
- II. Width: 64 cm
- III. Thickness: 0.04 cm

3.2.3. Perforated Pipe (part)

The perforated pipe, often referred to as the air rater, plays a vital role in creating a bubbling effect within the saline water. Its dimensions include:

- I. Diameter: 1cm
- II. Length: 180 cm

3.2.4. Side View of the Solar Still

- I. Width: 0.6 m
- II. Back side wall height: 0.27 m
- III. Front side wall height: 0.06 m
- IV. Specific height: 0.165 m
- V. Slope angle: 19°

3.2.5. Final Assembly

The final assembly combines all elements of the solar desalination system into a cohesive unit, featuring the following specifications:

- I. Length: 1.2 m
- II. Width: 0.6 m
- III. Back side wall height: 0.27 m
- IV. Front side wall height: 0.06 m
- V. Specific height: 0.165 m
- VI. Slope angle: 19°
- VII. Basin aspect ratio: 2
- VIII. Glass thickness: 0.004 m

These dimensions are carefully designed to optimize the functionality and efficiency of the stills, ensuring effective water production in varying environmental conditions.

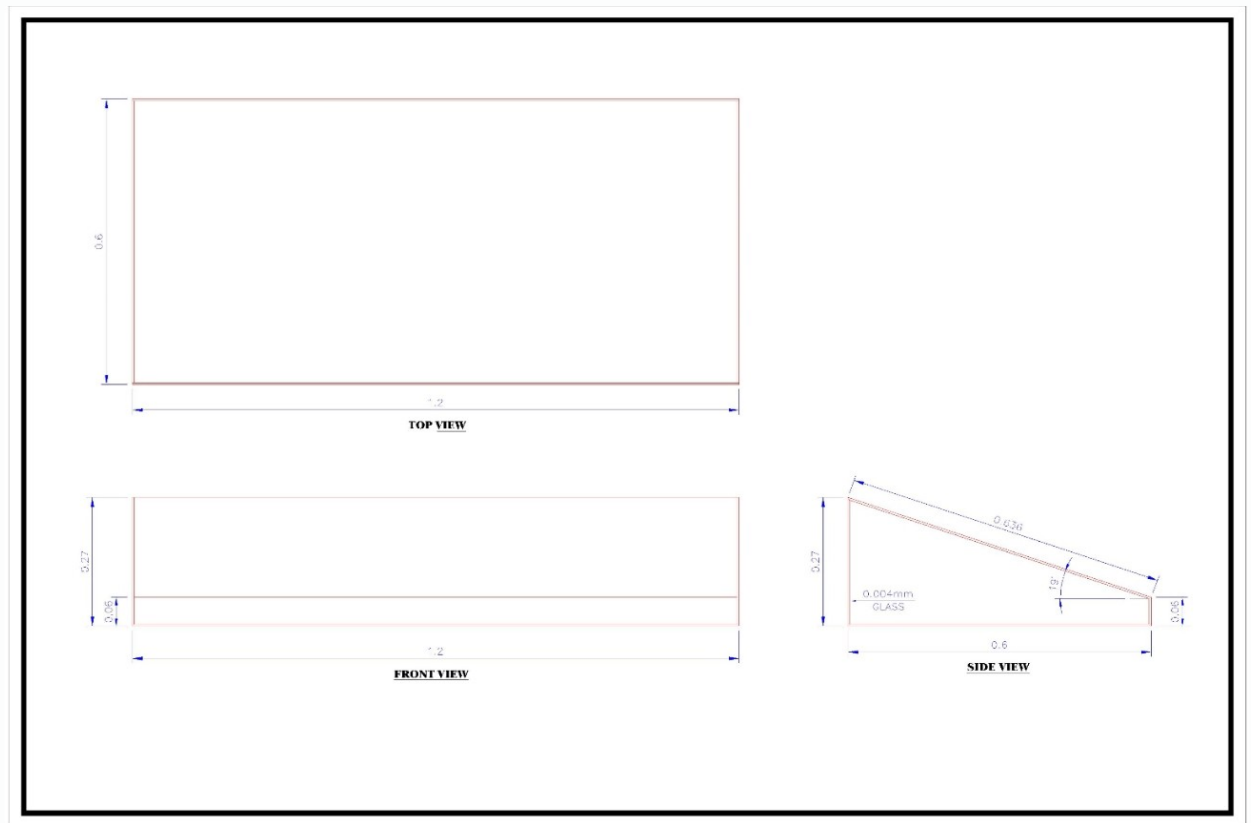


Figure 3: Shows the 2-D view of the single slope solar still.



Figure 4: Shows the CAD model of solar still without Aerator.

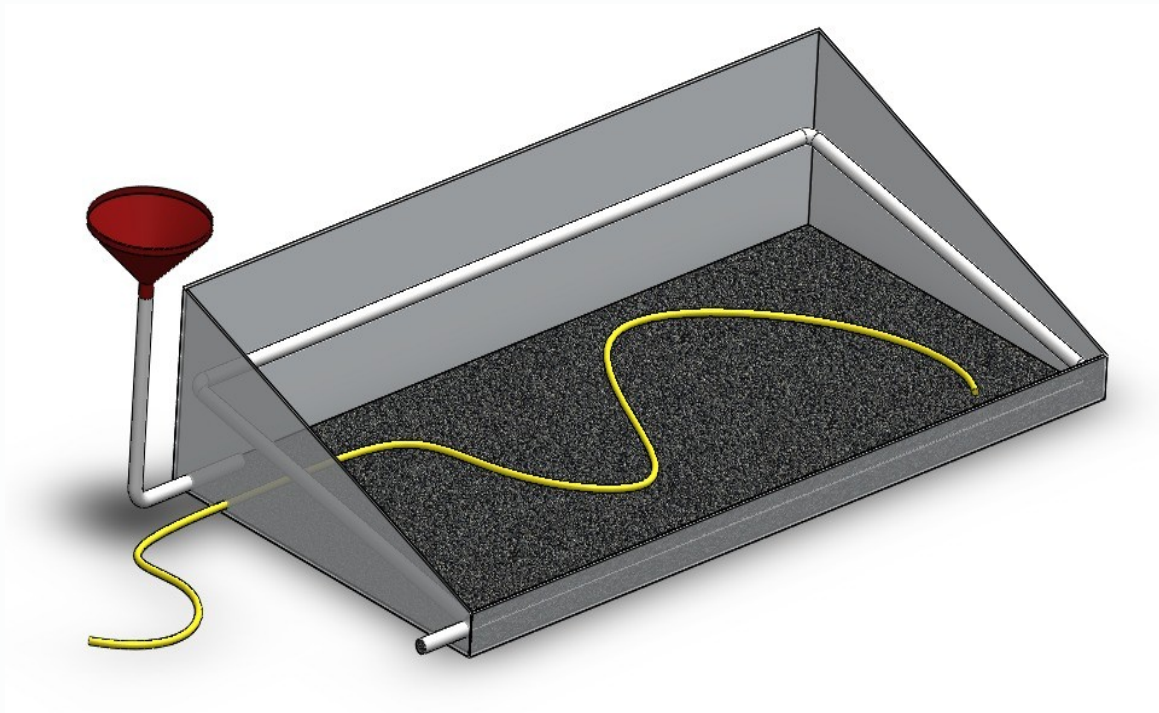


Figure 5: Shows the CAD model of solar still with Aerator.

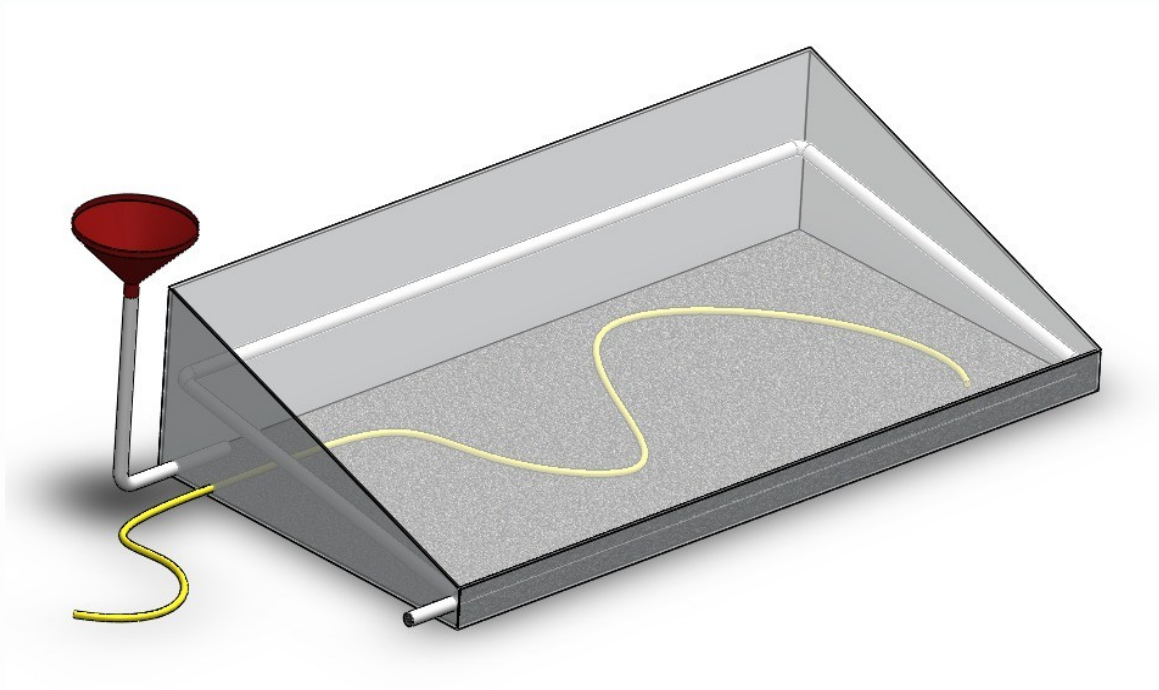


Figure 6: Shows the Final Assembly of Solar Still with Aerator and Upper Glass Cover.

3.3. FABRICATION

The fabrication process involved in building a solar desalination system. Every component, from the absorbent plate to the final assembly, undergoes careful fabrication to ensure system functionality and durability.

3.3.1. Aerator and PVC pipe:

Making the aerator and PVC pipes involves several steps. First, the PVC pipes are cut to the required length and ready for installation. After that, the air supply parts, including the fan and diffuser, are assembled and firmly attached to the PVC pipes. Correct alignment and sealing of seams must be ensured to prevent air or water leakage shown below in fig 7.

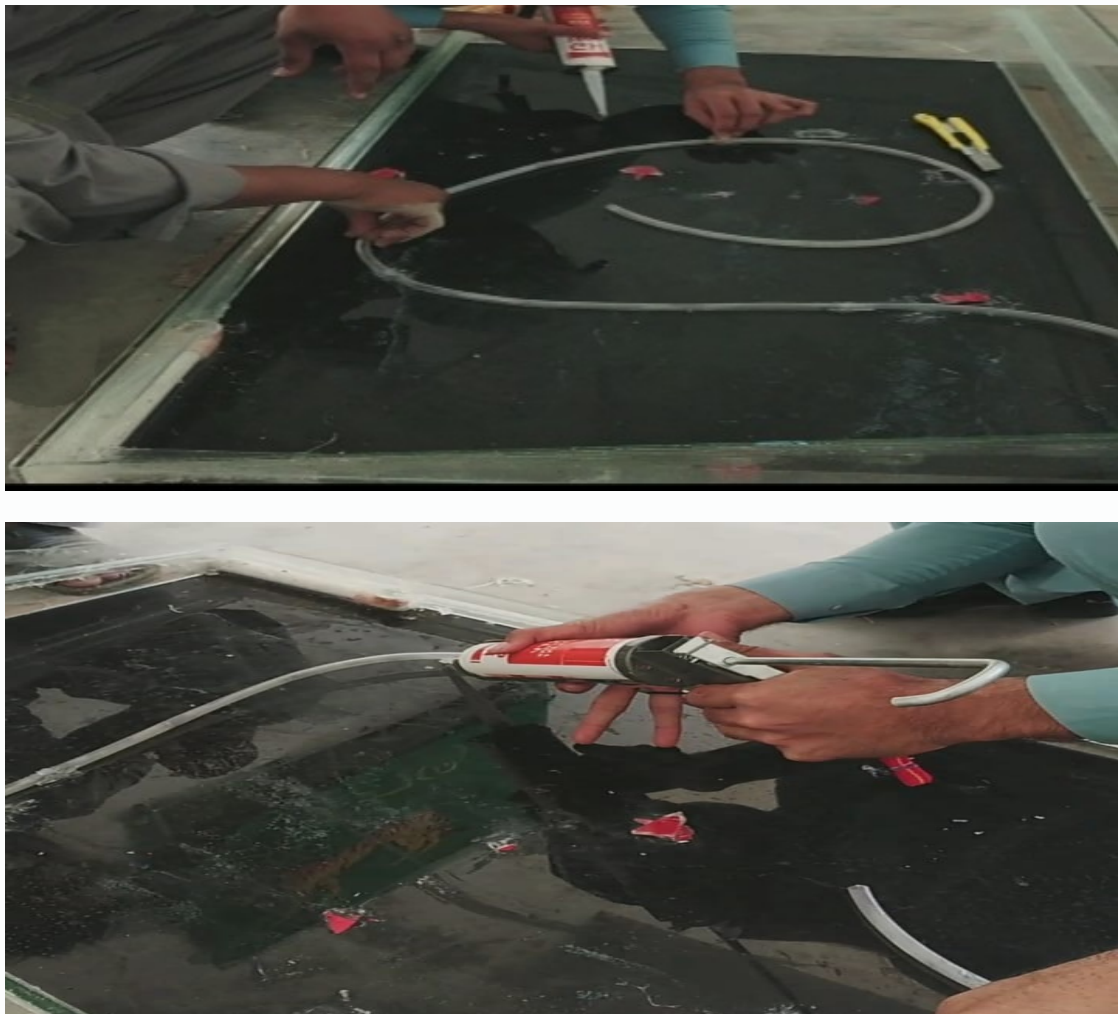


Figure 7:Shows the Fabrication of Perforated Tube

3.3.2. Ceramic base:

Making a ceramic base starts with choosing a high-quality ceramic material that meets the thermal and structural requirements of the system. The ceramic material is molded and then molded into the solar switch base. Precision cutting and finishing techniques are used to achieve the desired dimensions and surface smoothness. The ceramic base is hardened and tempered to improve its strength and durability and backend to improve the absorptivity shown below in fig 8.



Figure 8: Shows the Fabrication of Ceramic Base

3.3.3. Glass cover:

Making a glass cover requires careful handling of the glass material to avoid breakage and to ensure uniformity of thickness and transparency. Sealing the edges with silicone glue or similar materials ensures a waterproof and durable case as shown in fig 9.



Figure 9: Fabrication of Glass Cover.

3.3.4. Glass wall cutting:

Glass wall cutting involves precision cutting techniques to create openings for entrances and outlets. Special glass tools are used to achieve clean and precise cuts without compromising the structural integrity of the glass. Careful measurements and markings are made to ensure proper alignment and compatibility with other system components as shown in fig 10.



Figure 10: Fabrication of Glass Wall cutting.

3.3.5. Installation of PVC pipes:

Installation of PVC pipes requires their installation in predetermined locations in the solar desalination system. Correct alignment and spacing are critical to ensure optimal water and air flow through the system. The tubes are securely fastened with suitable fittings and glue to prevent leakage or loosening during use as shown in fig 11.



Figure 11: Fabrication of PVC Pipe Installation

3.3.6. Installation of Accessories:

Various accessories such as thermal sensors, flow gauge and pressure meters are assembled in the design to monitor and control the most important parameters. These accessories are strategically placed to ensure accurate readings and facilitate real-time adjustments for optimal system performance as shown in fig 12.



Figure 12: Fabrication of Accessories

3.3.7. Assembly and Sealing:

Final assembly of a solar desalination system involves joining all manufactured components into a single unit. The components are carefully placed and are fixed with glue. Sealing joints and joints with silicone watertight and airtight enclosure, minimizing the risk of leakage and maximizing system efficiency. Careful control measures are carried out throughout the fabrication process to ensure accuracy, design integrity and functional performance of each component. Regular inspections and testing help to quickly identify and correct any problems and ensure that the final product meets the desired specifications and standards as shown in fig 13.

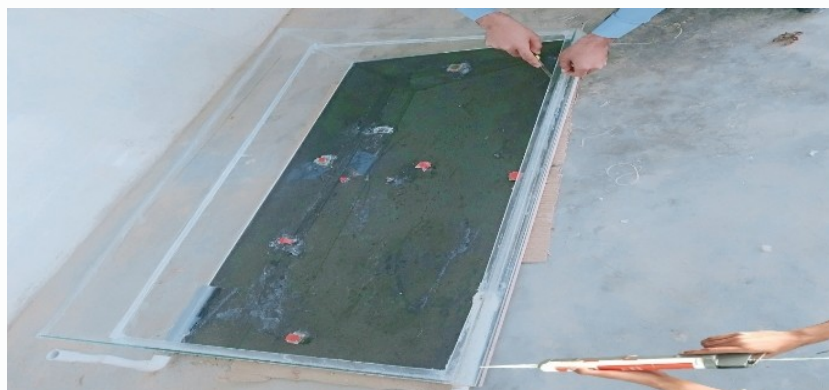


Figure 13: Assembly and Sealing

Mathematical Model

Chapter 4

Energy-balance formulation for the glass cover, basin water, liner and absorber plate, including distillate yield and overall efficiency.



CHAPTER 4.

4.1. MATHEMATICAL MODEL

In this chapter we delve into the mathematical formulation of a solar thermal desalination system. A numerical model is essential to investigate the efficiency, effectiveness and productivity of the solar powered desalination system. This model contains the equations of energy balance of each components of the system, the differential equations describing the heat flow and mass transport processes, and the solutions of these equations. Derived equations are validated against existing research data, ensuring their reliability and accuracy.

4.1.1. Glass cover:

The glass covering is a crucial component of the desalination system breaker that permits solar radiation to enter and minimizes heat loss. The energy balance equation of the glass covering can be written as follows: [23]

$$Q_i = Q_{out}$$

Where:

- (Q_i) is the incoming solar radiation.
- (Q_{out}) is the convective and radiative heat loss.

It can be modeled as follows:

$$Q_{gc} = h_{gc} A_{gc} (T_{gc} - T_a) + \epsilon_{gc} \sigma (T_{gc}^4 - T_{sky}^4)$$

Where:

- (Q_{gc}) is transfer heat through the glass covering
- (h_{gc}) is the convective coefficient of heat transfer in the glass covering.
- (A_{gc}) is the area of the glass covering.
- (T_{gc}) is the temperature of the glass cover.
- (T_a) is the ambient temperature.
- (σ) is the Stefan-Boltzmann constant.
- (ϵ_{gc}) means emissivity of the glass cover.
- (T_{sky}) means sky temperature.

4.1.2. Heat transfer in Basin water:

The energy balance equation for water in the basin can be expressed as: et.al. Verma, S., Das, R. [24]

$$m_w c_w \frac{dT_w}{dt} = Q_{\text{in}} - Q_{\text{evap}} - Q_{\text{conv}} - Q_{\text{rad}}$$

Where:

- (m_w) is the water's mass in the basin.
- (c_w) means specific heat capacity of water.
- ($\frac{dT_w}{dt}$) means rate of change of water temperature.
- (Q_{in}) is the absorbed solar radiation by water.
- (Q_{evap}) Is the evaporative heat loss.
- (Q_{conv}) is the heat loss from the water surface through convection.
- (Q_{rad}) is heat loss from the water surface through radiation.

4.1.3. Basin Liner:

The basin liner absorb energy from sun and heat up the water, so the energy balance equation is: Hu, Zheng, et al. [25]

$$Q_{bl} = \alpha_{bl} I A_{bl}$$

Where:

- (Q_{bl}) means the absorbed heat by basin liner.
- (α_{bl}) means the absorptivity of the basin liner.
- (I) means the intensity of light.
- (A_{bl}) means the area of the basin liner.

4.1.4. Evaporative heat loss:

The evaporative heat loss of the saline water, which is expressed in term of equation as: Wang, Lu, et al. [26]

$$Q_{evap} = m_{evap} h_{fg}$$

Where:

(Q_{evap}) is the evaporative heat loss.

(m_{evap}) is the mass of the water evaporated.

(h_{fg}) is the latent heat of vaporization.

4.1.5. Heat loss from water in the basin through convection:

The heat loss from the basin through convection to the surrounding air is calculated as follows:

Al-Kayiem, et.al. [27]

$$Q_{conv} = h_{cw} A_w (T_w - T_a)$$

Where:

- (Q_{conv}) is heat loss from basin water by convection.
- (h_{cw}) is convective heat transfer coefficient for water.
- (A_w) is the surface area of the water.
- (T_w) is the water's temperature.
- (T_a) Is the ambient temperature.

4.1.6. Heat loss from basin water through radiative:

Heat loss of basin water through radiation is calculated as follows: et.al. Verma, S., & Das, R. [28]

$$Q_{rad} = \sigma_w A_w (T_w^4 - T_{sky}^4)$$

Where:

- (σ) is the Stefan-Boltzmann constant.
- (σ_w) is the surface emissivity of the water.
- (A_w) shows surface area of the water.
- (T_w) means temperature of water.
- (T_{sky}) is the sky temperature.

4.1.7. Heat transfer in the Basin liner:

The energy balance equation of the basin trapping solar radiation and transform it to the water can be expressed as follows: [24]

$$m_{bl} c_{bl} \frac{dT_{bl}}{dt} = Q_{i\ell} - Q_{cond}$$

Where:

- (m_{bl}) shows the mass of the basin liner.
- (c_{bl}) shows the specific heat capacity of the basin liner material.
- $(\frac{dT_{bl}}{dt})$ is the rate of change in temperature of the basin liner.
- $(Q_{i\ell})$ is the radiation from sun absorbed by the basin
- (Q_{cond}) shows the conductive heat transfer from basin to water.

4.1.8. Heat transfer from basin to water through Conduction:

Heat transfer from basin to water through conduction is expressed as follows: da Silva, Karyna Steffane, et al. [29]

$$Q_{cond} = K_{bl} A_{bl} \left(\frac{T_{bl} - T_w}{d} \right)$$

Where:

- (Q_{cond}) shows conductive transfer of heat from basin liner to water
- (K_{bl}) shows device thermal conductivity of the basin liner material
- (T_{bl}) Shows the temperature of the basin liner.
- (T_w) Shows the temperature of water .
- (d) Is the thickness of the basin.

4.1.9. Heat transfer in the absorption plate:

The absorption plate plays a crucial role in heat collection and transfer. The heat balance equation of the absorber plate is given by the formula: Verma, S., & Das, R. et.al. [28]

$$m_{ap} C_{ap} \frac{dT_{ap}}{dt} = Q_{solar} - Q_{cond} - Q_{loss}$$

Where:

- (m_{ap}) Shows mass of the absorber plate.
- (C_{ap}) Shows the specific heat capacity of the absorbent plate material.
- ($\frac{dT_{ap}}{dt}$) is the rate of change of absorber plate temperature.
- (Q_{solar}) is the solar heat absorbed by the absorber.
- (Q_{cond}) shows the heat transfer to water through conduction.
- (Q_{loss}) is heat loss to the environment from the absorber plate.

4.1.10. Distillate per hour per unit area of the basin:

The hourly distilled is calculated according to the per unit basin area as: Alshqirate, AbedAlrzaq, et al [30]

$$D = \frac{Q_{evap}}{h_{fg} A_b}$$

Where:

- (D) Is the distillate yield per unit basin area
- (A_b) is the area of the basin.

4.1.11. Daily distillate yield:

The daily distillation of water is obtained by integrating the hourly distillation yield over 24 hours. Alshqirate, AbedAlrzaq, et al [30]

$$Q_{daily} = \int_0^{24} D dt$$

Where:

- (Q_{daily}) is the daily distilled water production.
- ($\int_0^{24} D dt$) is the integration operation of hourly distillation yield.

4.1.12. Overall Efficiency of solar stills:

The overall efficiency of the solar desalination system is defined as the ratio of the useful heat used for evaporation to the total solar energy production. Al-Kayiem .et.al [31]

$$\eta_{overall} = \frac{m_{evap} h_{fg}}{I A_b}$$

Where:

($\eta_{overall}$) is the overall efficiency.

(m_{evap}) means the mass of evaporated water.

(h_{fg}) is the specific heat of vaporization.

(I) shows the intensity of solar radiation.

(A_b) is the Basin area.

4.1.13. Conclusion:

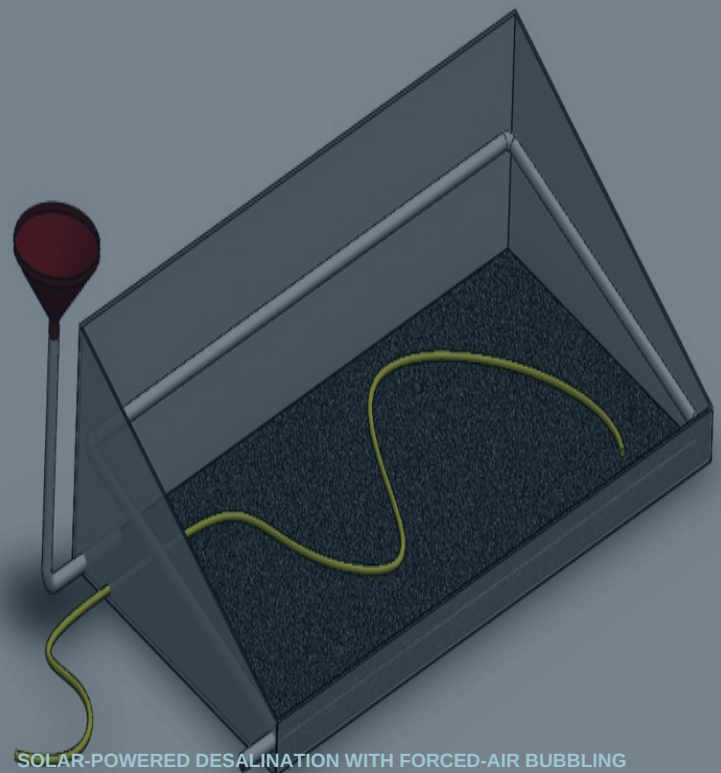
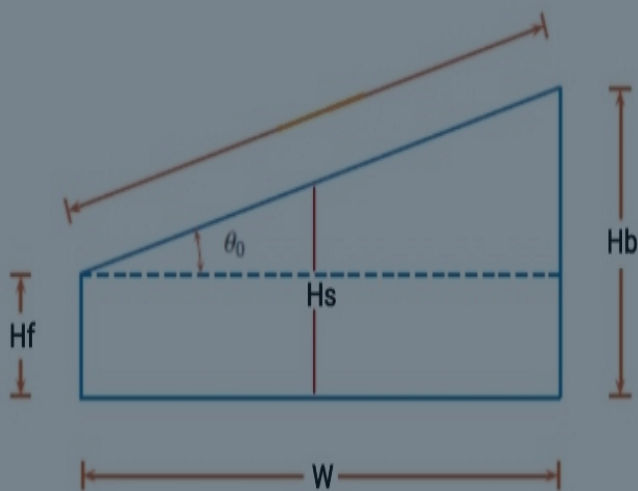
This section presents the formulation of a thorough mathematical modeling for the solar-powered desalination system, with particular emphasis on important parts like the glass covering, mass of water and basin. Through the derivation and solution of energy balance equations, we offered a thorough framework for forecasting the thermal behavior and distillate yield of the system; these equations, when verified against empirical data, allow precise simulations of the desalination process, directing the optimization of system design and performance. This analytical work not only advances our knowledge of the heat energy and mass transfer principles but also opens the door for future advances in solar desalination technology.

Market Survey & Procurement

Chapter 5

Selection and procurement of glass, aerator, piping, ceramic base, instruments and sealing materials.

(b)



CHAPTER 5.

5.1. MARKET SURVY AND PROCUREMENT

5.1.1. Glass cover:

Glass cover is the main component of a solar desalination system and acts as a transparent envelope that captures solar radiation and facilitates the evaporation process. In our market research, we explored different options for purchasing high-quality tempered glass suitable for use in solar energy applications. Thickness, transparency, durability and cost-effectiveness must be considered. Suppliers were evaluated based on product quality, price, delivery times and customer service. Finally, we chose a supplier that offers 4 mm tempered glass with optimal optical clarity and thermal insulation properties to meet our design requirements (Ahmed, Hamza, et al. [8])

5.1.2. Aerator:

The aerator or air regulator performs a critical role in increasing the evaporation rate in a solar desalination system by causing mixing and turbulence in the brine pool. In our market research, we identified suppliers that offer suitable perforated pipes or aerators to introduce air into the water. The factors evaluated were design, material composition, aerator efficiency and compatibility with our system requirements. After evaluating several options, we obtained a perforated tube with small holes to ensure uniform air distribution and optimal water mixing, which contributed to increased freshwater production. Isah, Abubakar Sadiq, et al. [32].

5.1.3. Pipe:

Pipes are important for transporting water, air and other fluids in a solar desalination system. In our market research, we identified suppliers that offer pipes for a variety of applications, including water inlet and outlet, air supply and drainage. Among other things, material composition, diameter, wall thickness and corrosion resistance were considered. After comparing options from several suppliers, we chose high-quality PVC pipes known for their durability, flexibility and chemical resistance, ensuring reliable performance and longevity of our system.

5.1.4. Ceramic base:

The ceramic base provides a stable base for the solar desalination system that supports the pool and other components. In our market research, we investigated suppliers that offer ceramic materials suitable for the substructure. Strength, thermal conductivity, chemical resistance and price should be considered. After evaluating several options, we chose a ceramic material known for its durability, heat retention and suitability for outdoor use. The selected ceramic base provides a strong foundation for the system, while it can withstand the harsh demands of use in various environmental conditions.

5.1.5. Instruments:

Various tools and equipment are required to assemble, install and maintain a solar desalination system. During our market research, we identified suppliers that offer essential instruments such as flintlock guns, knives, glass cutters, scissors and other hand tools. Quality, durability, ergonomics and cost-effectiveness had to be considered. After evaluating several supplier options, we acquired a complete set of high-quality equipment that meets our project requirements and ensures efficient and safe operation throughout the project life cycle (Ahmed, Hamza, et al [8]).

5.1.6. Silicone adhesive:

Silicone adhesive is used to seal joints and secure components to a solar desalination system to prevent leaks and ensure structural integrity. During market research, we investigate suppliers that offer silicone adhesives suitable for gluing glass, ceramics and other materials exposed to high temperatures and harsh environments. Adhesion strength, cure time, temperature tolerance and compatibility with project materials were taken into account. After careful evaluation, we selected a high-performance silicone adhesive known for its excellent adhesions, durability and UV and weather resistance, ensuring long-term reliability and performance of the system (Ahmed, Hamza, et al. [8]).

Table 1: Shows the total cost of both the Systems.

DESALINATION 2.3 LITERS/DAY 60 cm		X ² = 2.75 LITERS/DAY		
120 cm		60 cm	0.72 SQ meter	
SYSTEM ITEMS FOR 2.7 LITERS/DAY		QUANTITY	COST	TOTAL
COST/BASE 120 60 cm(RS4,000 /BASE) QTY TWO		2	3,500	7,000
GLASSES(RS300/SQFT)		28	300	8,400
BASE INSULATION POLY CARBONAT SHEETS 2 (2FT 4FT)		2	2,400	4,800
UV FILM 2FT 4FT(RS30/SQFT)		1	1,000	1,000
PIPES(HALF INCH 6FET RS 15/SQFT)		6	15	90
FRIDGE BEND 24(RS 40/PC)		4	65	260
TEE- ½ INCH(1)		4	30	120
ELBOWS-1/2 INCH		8	25	200
UNION ½ INCH		4	80	320
VALVES - ½ INCH (1 130/PC)		2	130	260
WATER TANK (USED RS 1300/PC)		2	1300	2,600
TEE CROSS – ½ INCH(1)		1	100	100
FLOAT VALVE – ½ INCH ONE RS (ONE RS 250/PC)		1	250	250
SUB TOTAL				25,400
EXPENDABLES				
SILICON BOTTLES(TWO RS240/PC)		2	1,000	2,000
ELPHIE (ONE RS35/BTL)		1	35	35
PIPE JOINING SOLUTION(RS440/CAN)		1	440	440
SUB TOTAL				2,475
LABOUR		1	2,000	2,000
TRANSPORTATION RS5,000 FOR SEVERAL TRIPS		1	7,000	7,000
SUB TOTAL				9,000
USER KITS ITEMS				
CLEANING KIT		1	50	50
WIPER		1	100	100
BROCHURES		2	100	200
STICKERS		2	500	1,000
SUB TOTAL				1,350
STANDARD COST				
Future Experiment to be Conducted for performance Enhancement				
AIR ELECTRIC REGULATED PUMP		1	1,500	2,097
BATTERY		1	3,000	3,000
BATTERY CHARGER		1	1,500	1,500
SUB TOTAL				6,000
GRAND TOTAL				44,822
COST WITH SALES TAX 17%		0.17	7,178	52,000

Testing

Chapter 6

Experimental preparation, operating procedure and performance indicators used during testing.



SOLAR-POWERED DESALINATION WITH FORCED-AIR BUBBLING

CHAPTER 6.

6.1. TESTING

The important step in the development is testing of a solar desalination system to investigate its functionality, efficiency and reliability in real-world conditions. This chapter describes the procedures and methods for comprehensive system testing as shown in fig 14.



Figure 14: Shows the Setting up Equipment.

6.1.1. Preparations for Test:

Before starting the test, we went through several preparatory steps to ensure that the system is ready and working. This includes filling the pool with salt water, adjusting the position of the solar switch to optimize exposure to sunlight, and filling the air metering system. Any air bubbles or impurities in the water are cleaned so that it does not affect the test results as shown in fig 15 & 16.



Figure 15: Checking the TDS of Inlet Water.

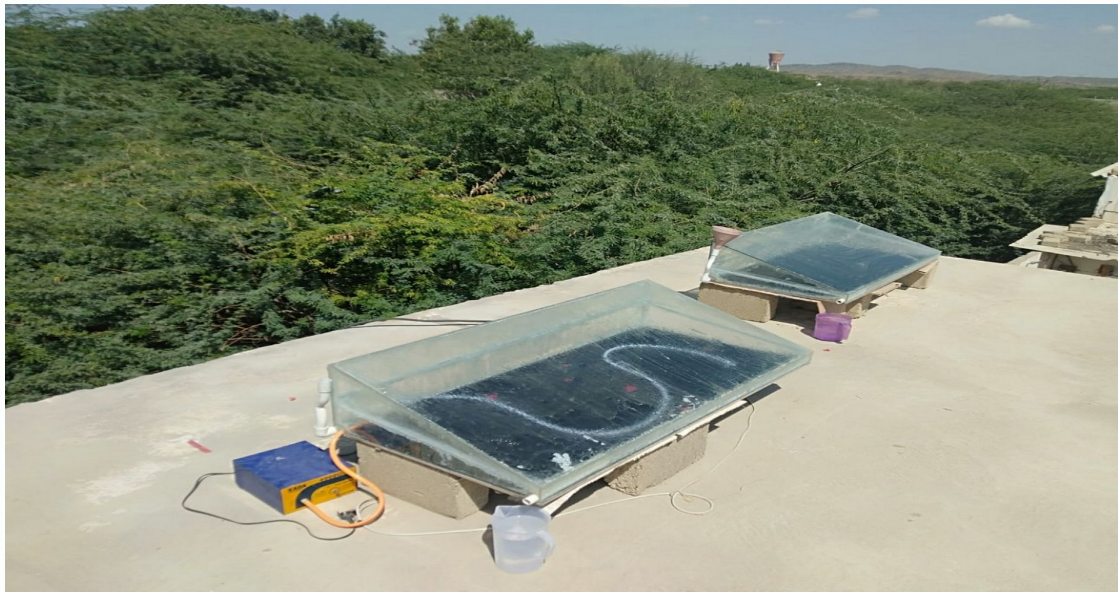


Figure 16: Shows Both Systems in running conditions.

6.1.2. Testing:

After finishing the preparations, we started the testing phase, during which the function of the system is evaluated under various weather conditions. This included measuring freshwater production rates, monitoring energy consumption and assessing water quality parameters such as salinity and pH. Testing is done to collect data for different time periods and environmental conditions as shown below in fig 17.



6.1.3 Several key performance indicators are evaluated during testing

a) Freshwater Production:

The freshwater production rate of the desalination system was measured and recorded over time. This gives an idea of the production and performance of the system to convert sea or brackish water into drinking water.

b) Efficiency of Energy:

Energy consumption and solar usage are clearly monitored to evaluate the energy performance of the system. This involved calculating the energy given required to drive a desalination process and determining the total energy conversion of the system.

c) Water quality:

Produced water quality has been analyzed to ensure that it meets prescribed drinking water standards. Parameters such as salinity, dissolved solids (TDS), turbidity and microbiological contamination were measured and compared with official guidelines.

d) System Reliability:

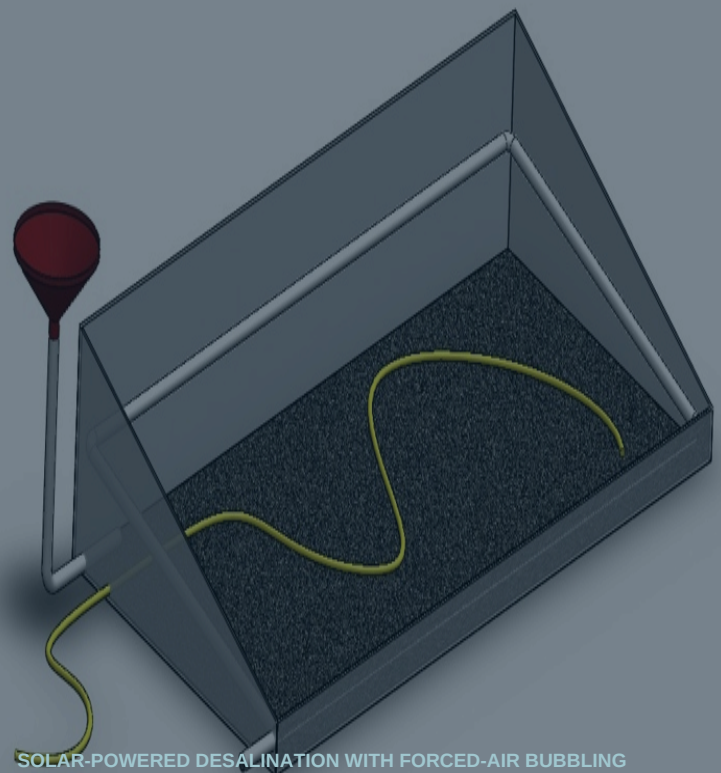
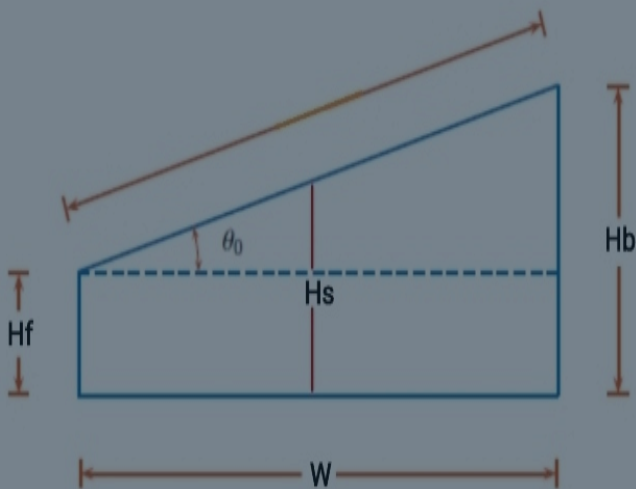
System reliability and robustness were evaluated by evaluating its performance under various operating conditions and stress test scenarios. This involved studying the feedback of the system to changes in solar radiation, ambient temperature and feed water composition. During the testing phase, data was continuously collected, monitored and analyzed to identify any anomalies or deviations from expected performance. Any problems or challenges encountered during testing were documented, and corrective actions were taken as needed to optimize system performance and reliability. Through rigorous testing of the solar desalination system, we confirmed its functionality, identified areas for improvement, and demonstrated that it is suitable for practical applications for solving water scarcity problems. Insights from the testing provided valuable feedback to improve system design, optimize operating parameters and improve overall performance.

Experimental & Theoretical Results

Chapter 7

Four months of experimental and theoretical results, including temperatures, solar intensity and freshwater productivity.

(b)



CHAPTER 7.

7.1. EXPERIMENTAL AND THEORETICAL RESULTS

This chapter shows the comprehensive results of the data analyzed during the experiments done throughout the four months and shows graphs of hourly temperature vs time, solar intensity vs time and distilled water production by system with aerator and without aerator.

[Date: 25-March-24].

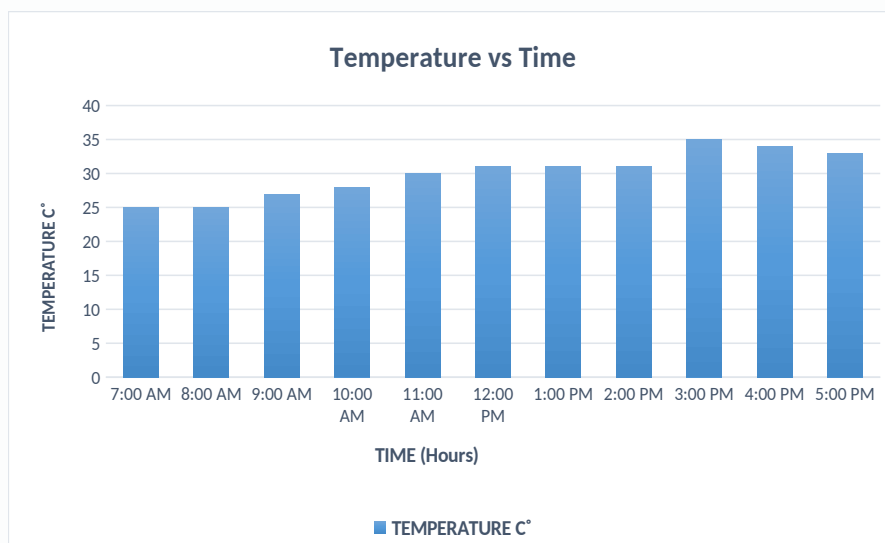


Figure 18: Shows the Hourly ambient temperature vs time of the [Date: 25-March-24].

[Date: 25-March-24]

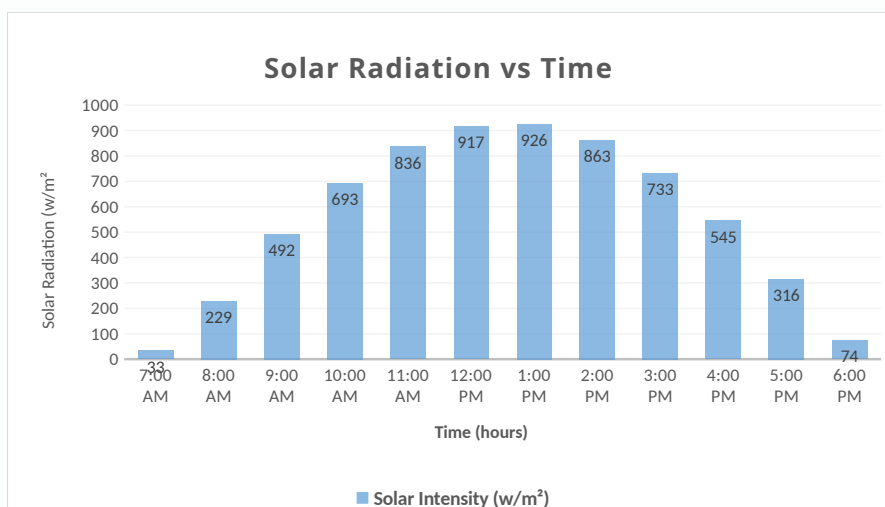


Figure 19: Shows the Hourly Solar Intensity vs Time of the [Date: 25-March-24]
[Date: 25-March-24].

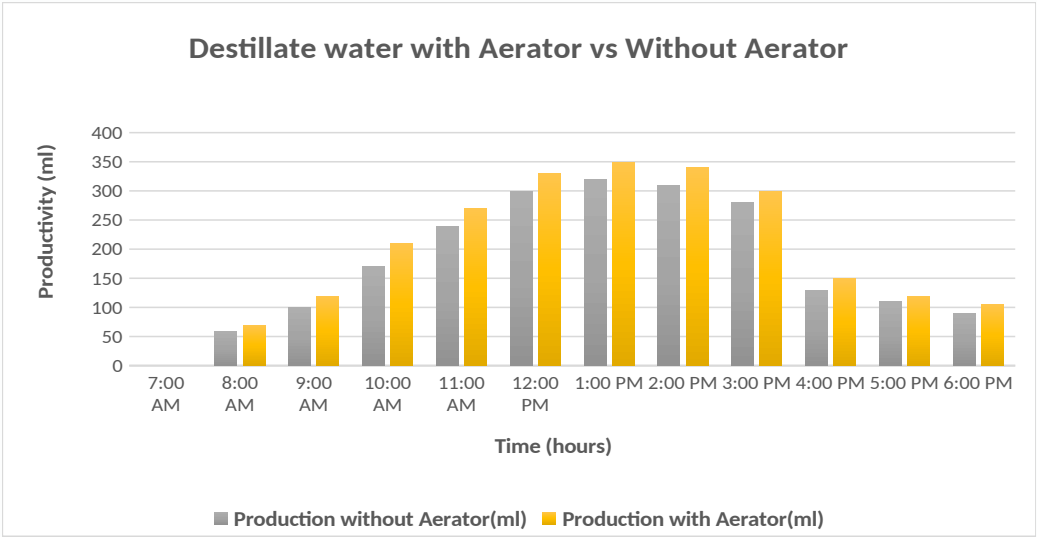


Figure 20: Shows the Hourly distillate water productivity by desalination with aerator vs without aerator of the [Date: 25-March-24].
[Date: 30-March-24].

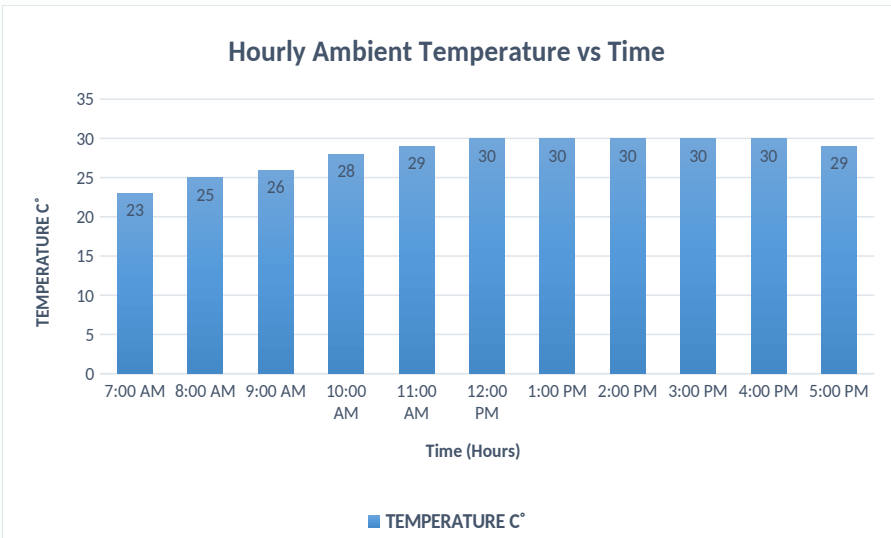


Figure 21: Shows the Hourly ambient temperature vs time of the [Date: 30-March-24].

[Date: 30-March-24].

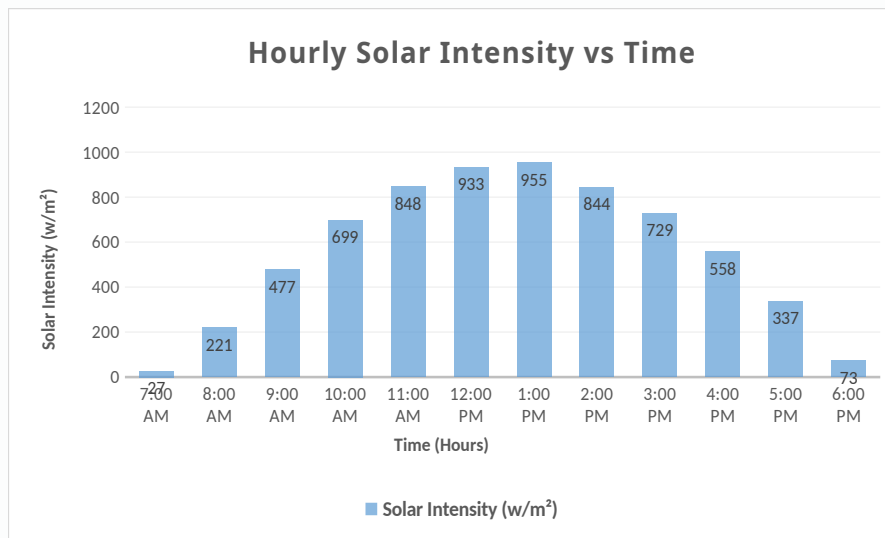


Figure 22: Shows the Hourly Solar Intensity vs Time of the [Date: 30-March-24].

[Date: 30-March-24].

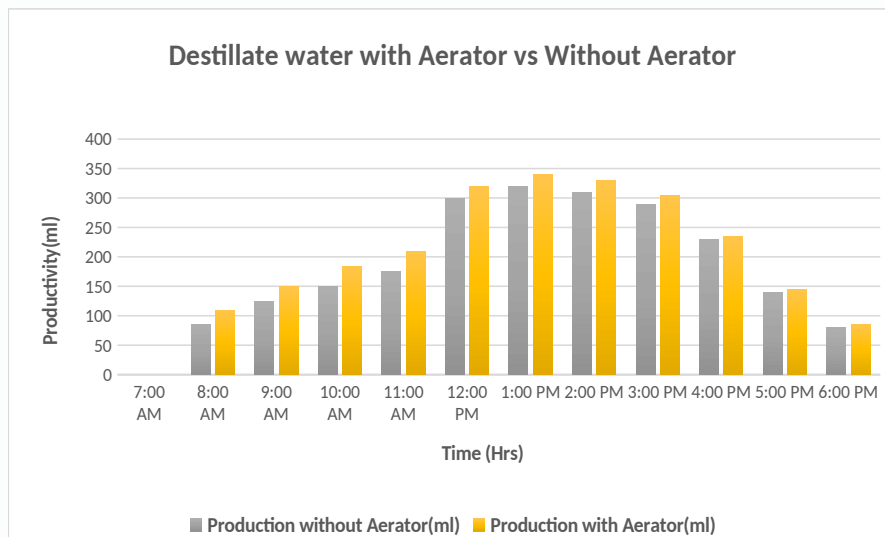


Figure 23: Shows the Hourly distillate water productivity by desalination with aerator vs without aerator of the [Date: 30-March-24].

[Date: 03-April-24].

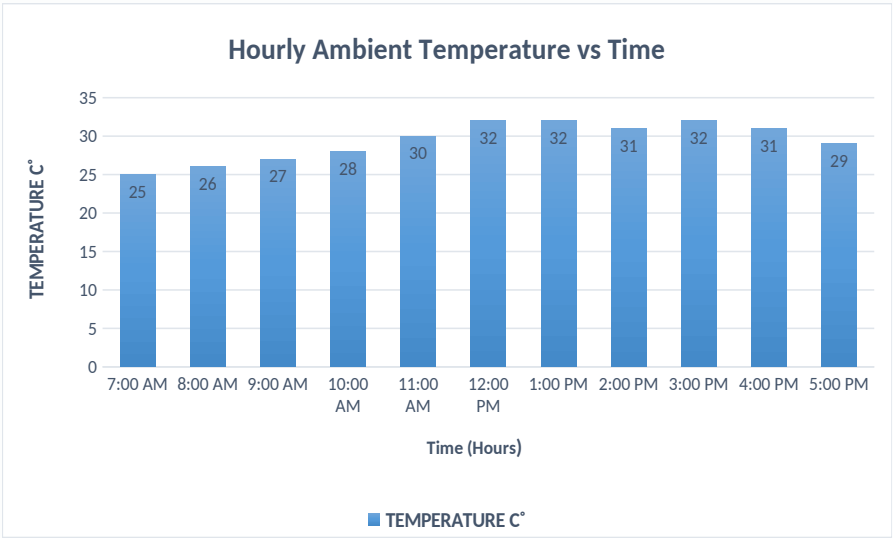


Figure 24: Shows the Hourly ambient temperature vs time of the [Date: 03-April-24].

[Date: 03-April-24].

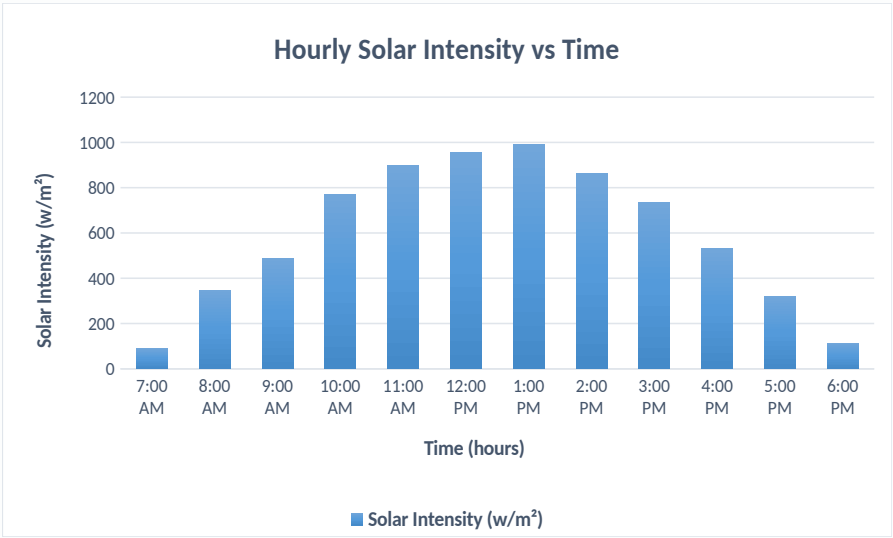


Figure 25: Shows the Hourly Solar Intensity vs Time of the [Date: 03-April-24].

[Date: 03-April-24].

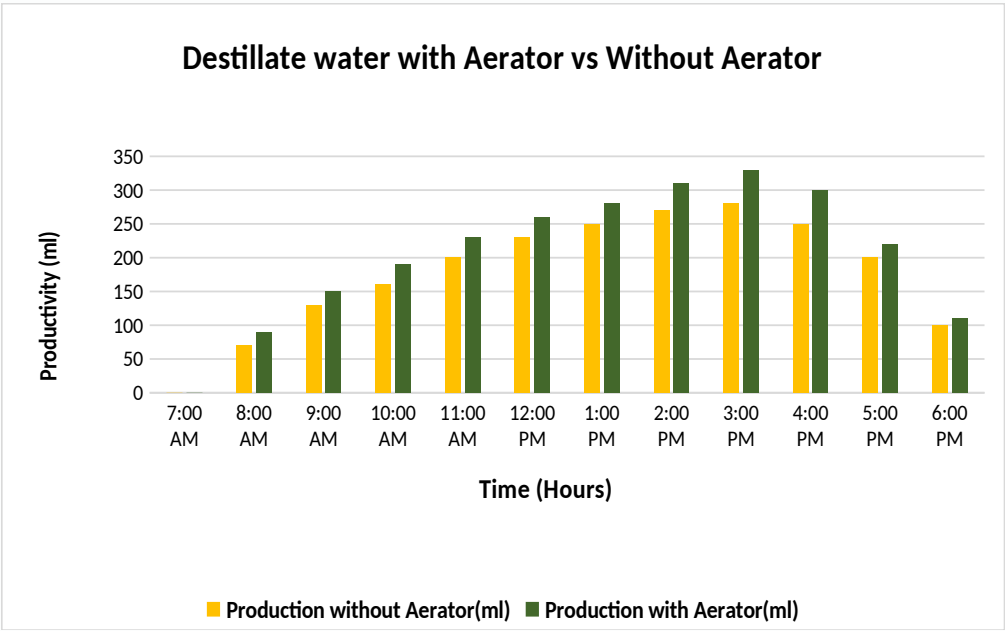


Figure 26: Shows the Hourly distillate water productivity by desalination with aerator vs without aerator of the [Date: 03-April-24].

[Date: 20-April-24].

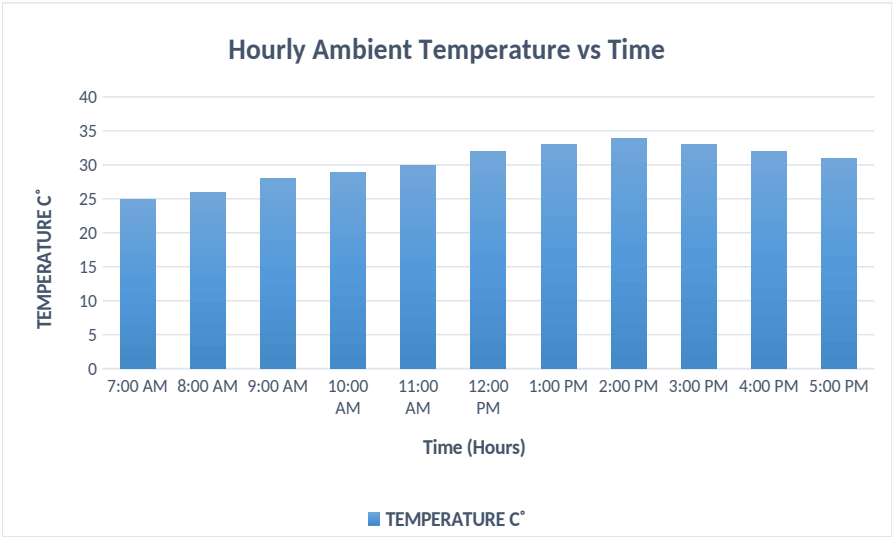


Figure 27: Shows the Hourly ambient temperature vs time of the [Date: 20-April-24].

[Date: 20-April-24].

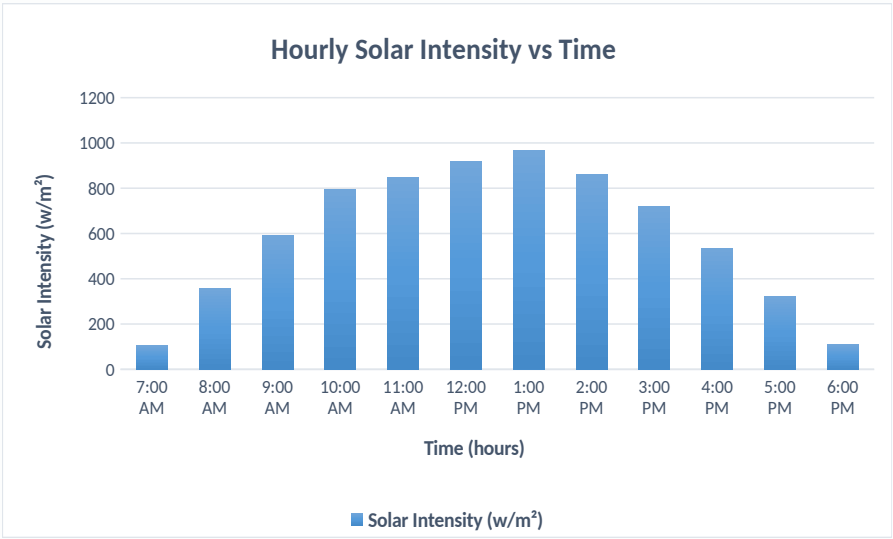


Figure 28: Shows the Hourly Solar Intensity vs Time of the [Date: 20-April-24].

[Date: 20-April-24].

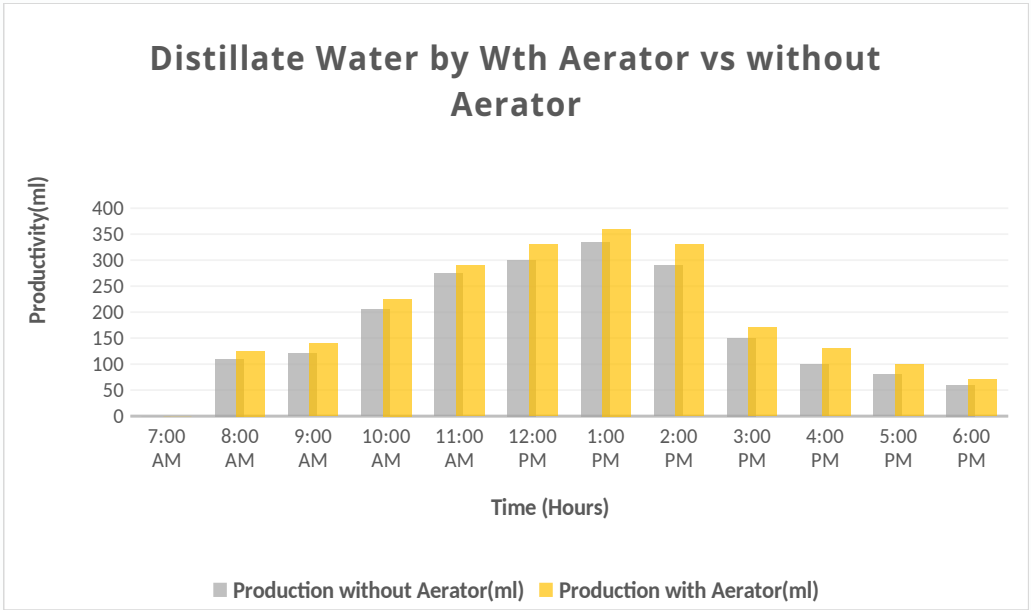


Figure 29: Shows the Hourly distillate water productivity by desalination with aerator vs without aerator of the [Date: 20-April-24].

[Date: 25-April-24].

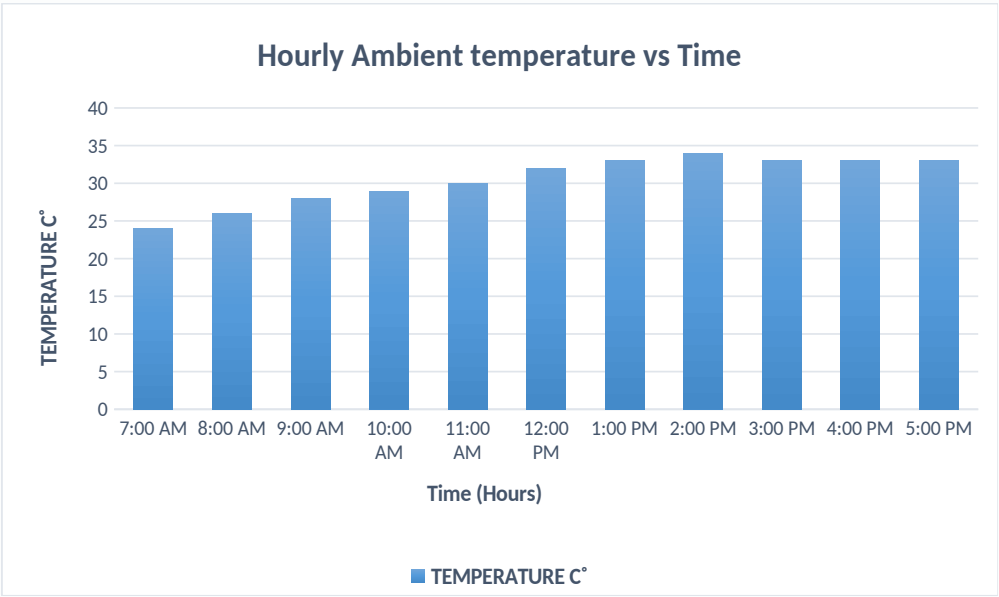


Figure 30: Shows the Hourly ambient temperature vs time of the [Date: 25-April-24].

[Date: 25-April-24].

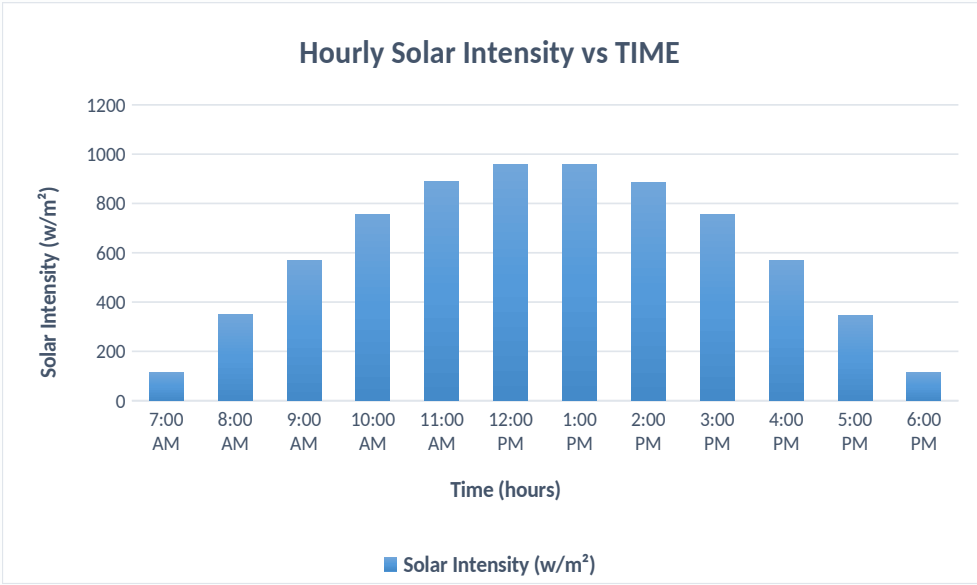


Figure 31: Shows the Hourly Solar Intensity vs Time of the [Date: 25-April-24].

[Date: 25-April-24].

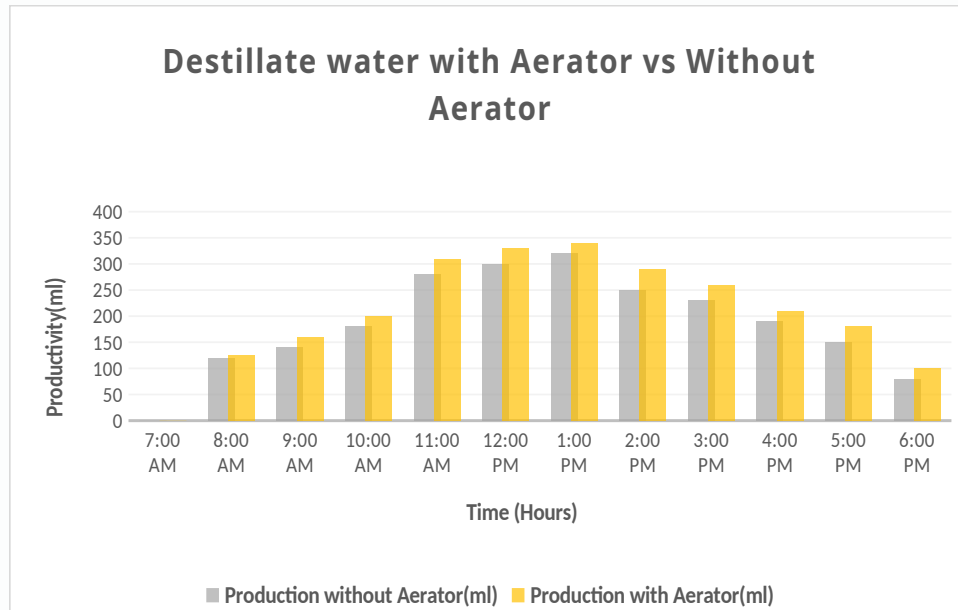


Figure 32: Shows the Hourly distillate water productivity by desalination with aerator vs without aerator of the [Date: 25-April-24].

[Date: 05-May-24].

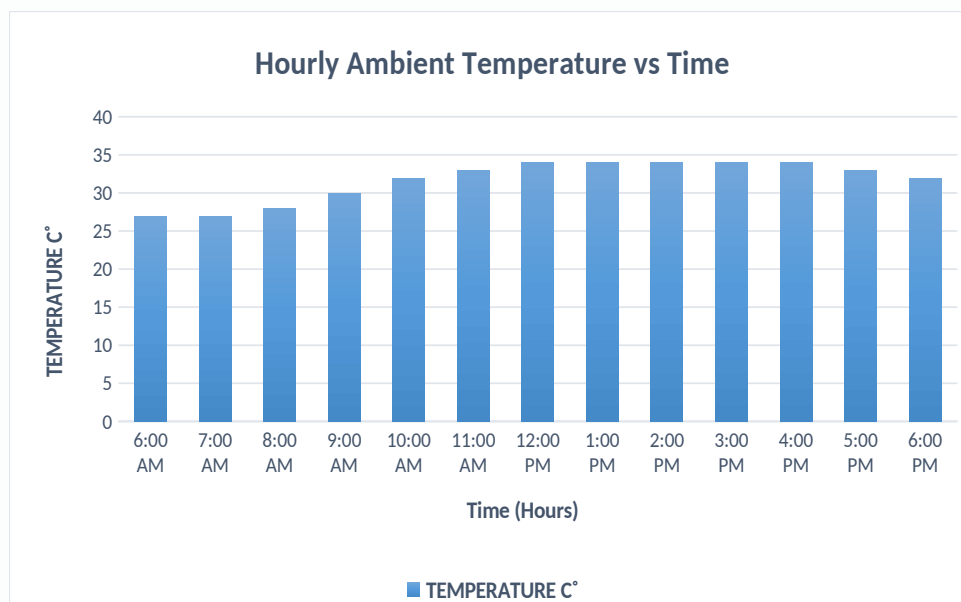


Figure 33: Shows the Hourly ambient temperature vs time of the [Date: 05-May-24].

[Date: 05-May-24].

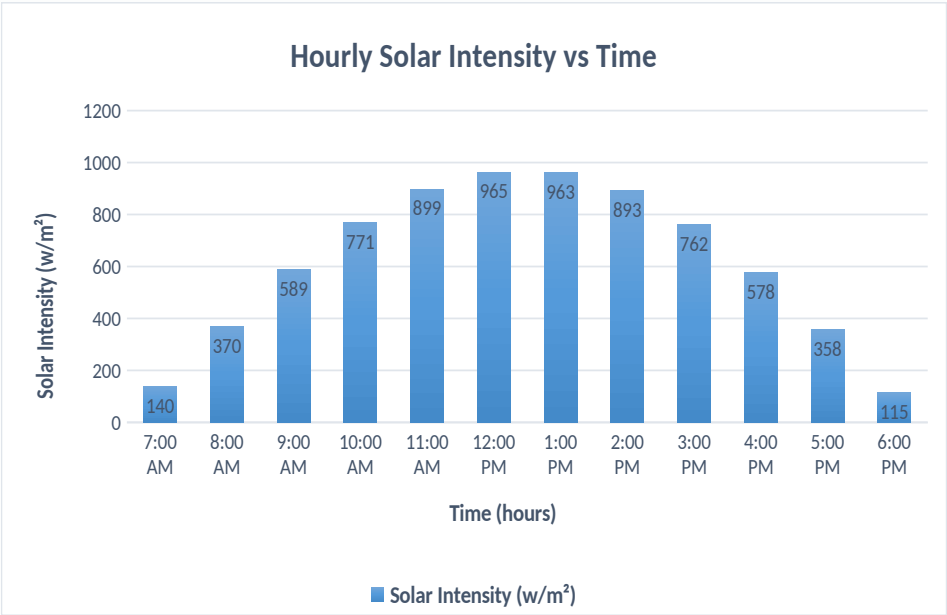


Figure 34: Shows the Hourly Solar Intensity vs Time of the [Date: 05-May-24].

[Date: 05-May-24].

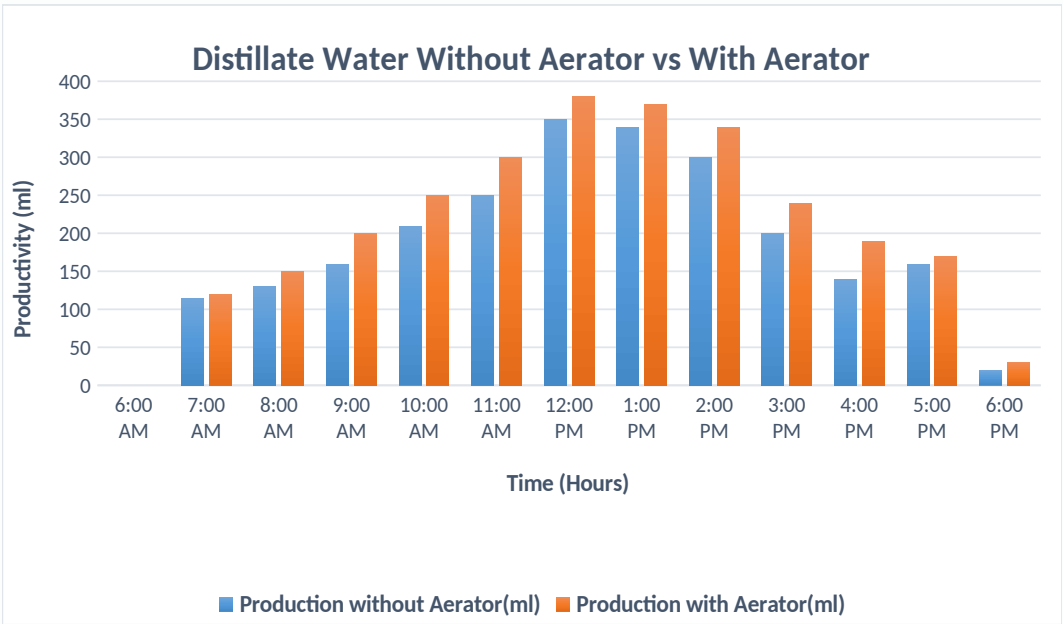


Figure 35: Shows the Hourly distillate water productivity by desalination with aerator vs without aerator of the [Date: 05-May-24].

[Date: 10-May-24].

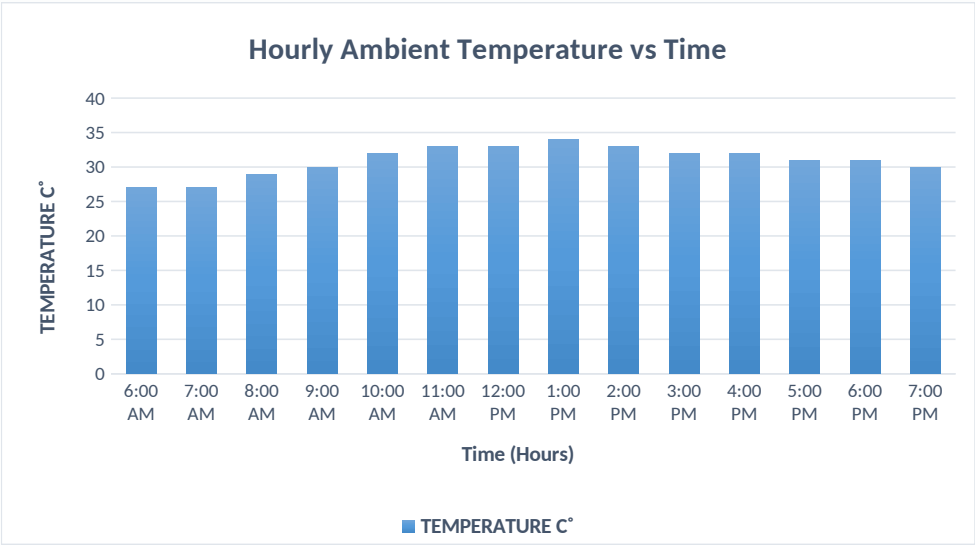


Figure 36: Shows the Hourly ambient temperature vs time of the [Date: 10-May-24].

[Date: 10-May-24].

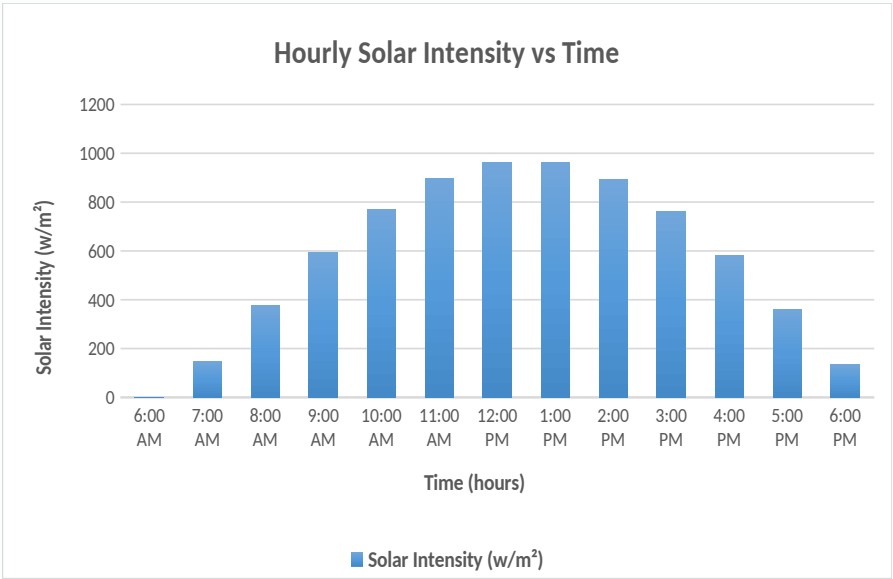


Figure 37: Shows the Hourly Solar Intensity vs Time of the [Date: 10-May-24].

[Date: 10-May-24].

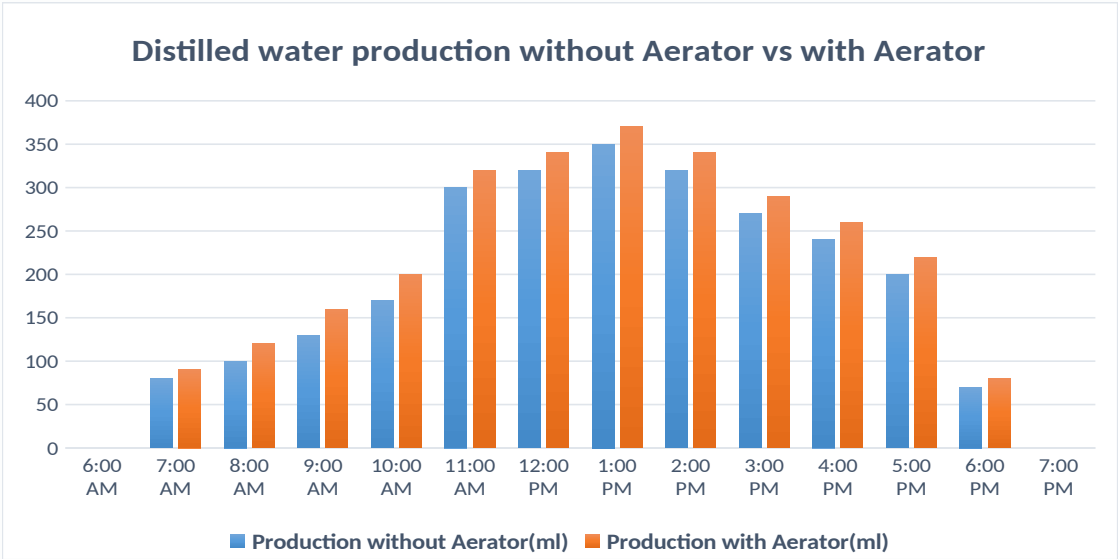


Figure 38: Shows the Hourly distillate water productivity by desalination with aerator vs without aerator of the [Date: 10-May-24].

[Date: 15-May-24].

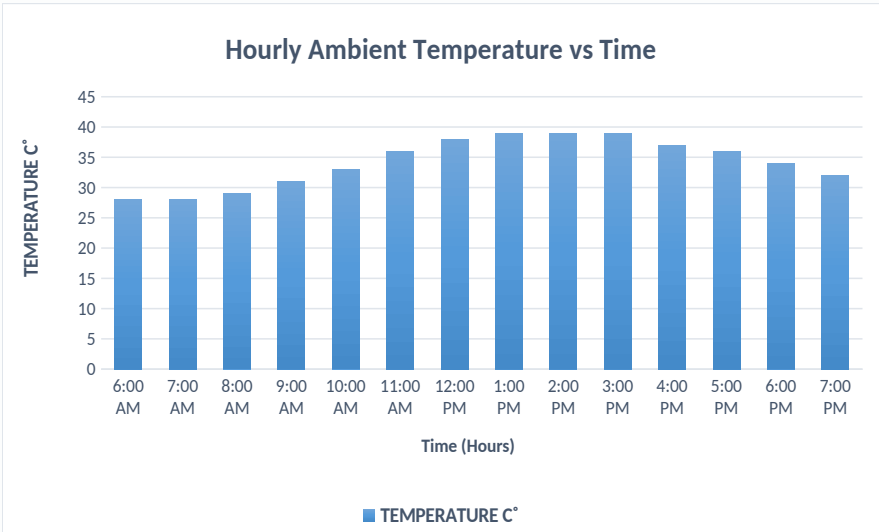


Figure 39: Shows the Hourly ambient temperature vs time of the [Date: 15-May-24].

[Date: 15-May-24].

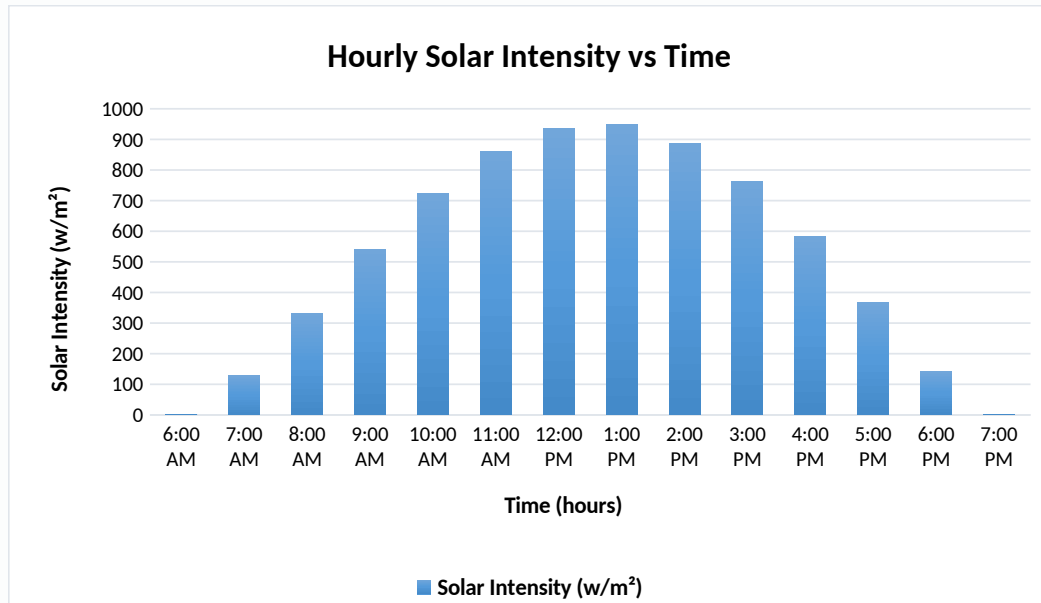


Figure 40: Shows the Hourly Solar Intensity vs Time of the [Date: 15-May-24].

[Date: 15-May-24].

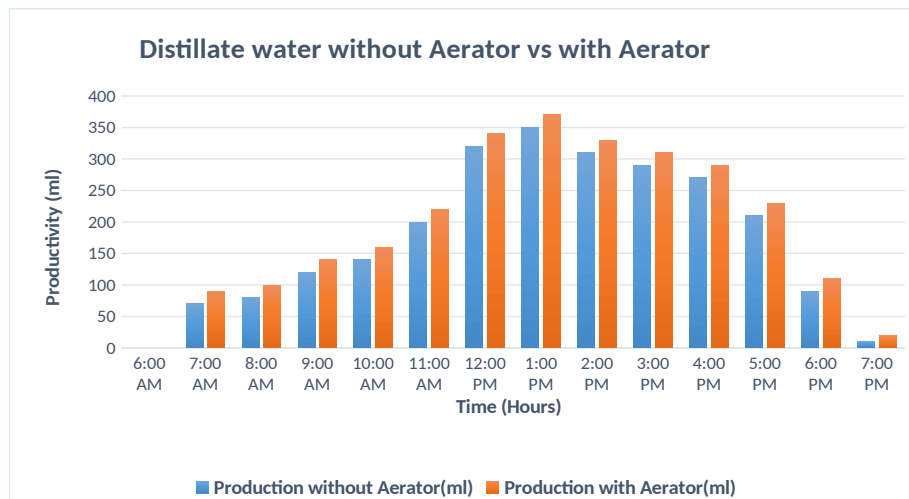


Figure 41: Shows the Hourly distillate water productivity by desalination with aerator vs without aerator of the [Date: 15-May-24].

[Date: 20-May-24].

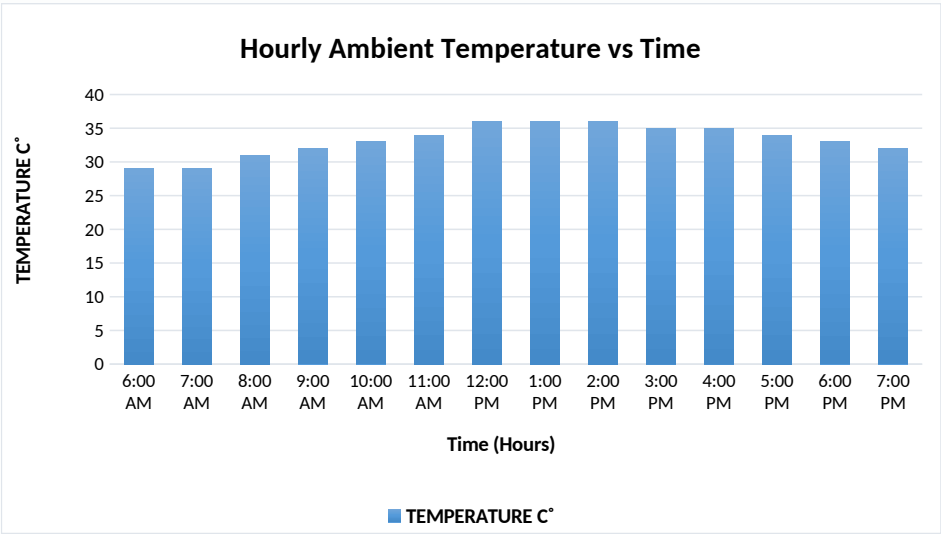


Figure 42: Shows the Hourly ambient temperature vs time of the [Date: 20-May-24].

[Date: 20-May-24].

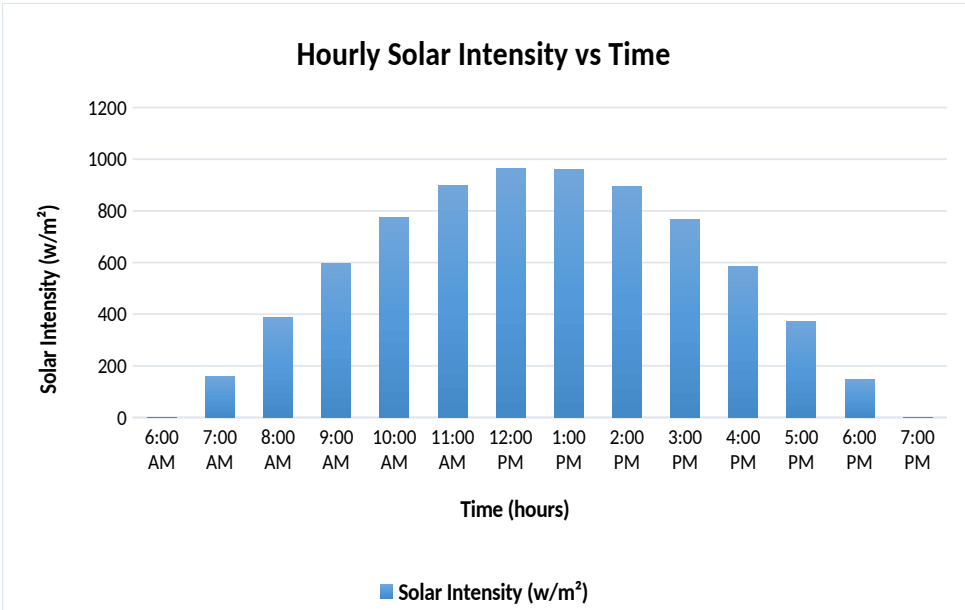


Figure 43: Shows the Hourly Solar Intensity vs Time of the [Date: 20-May-24].

[Date: 20-May-24].

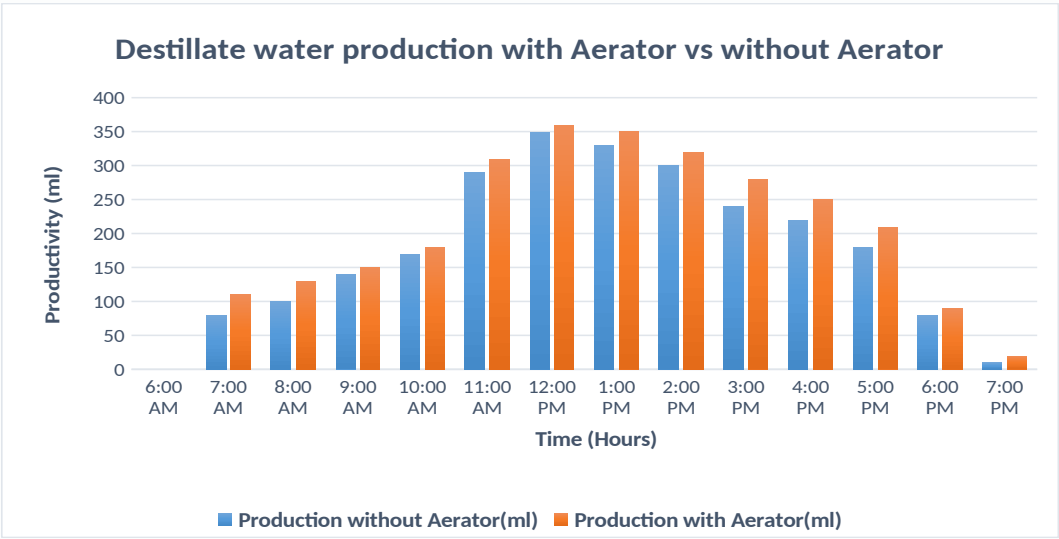


Figure 44: Shows the Hourly distillate water productivity by desalination with aerator vs without aerator of the [Date: 20-May-24].

[Date: 25-May-24].

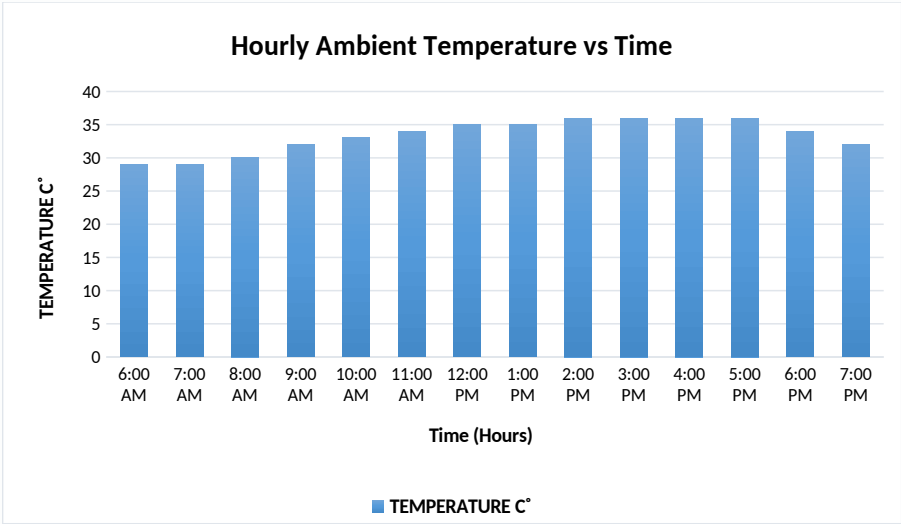


Figure 44: Shows the Hourly ambient temperature vs time of the [Date: 25-May-24].

[Date: 25-May-24].

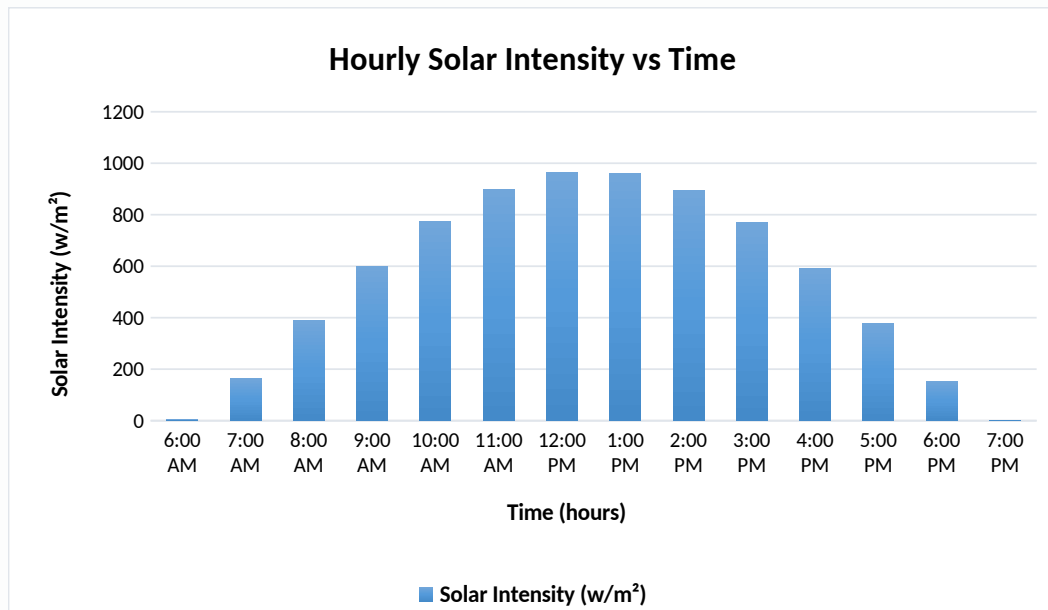


Figure 45: Shows the Hourly Solar Intensity vs Time of the [Date: 25-May-24].

[Date: 25-May-24].

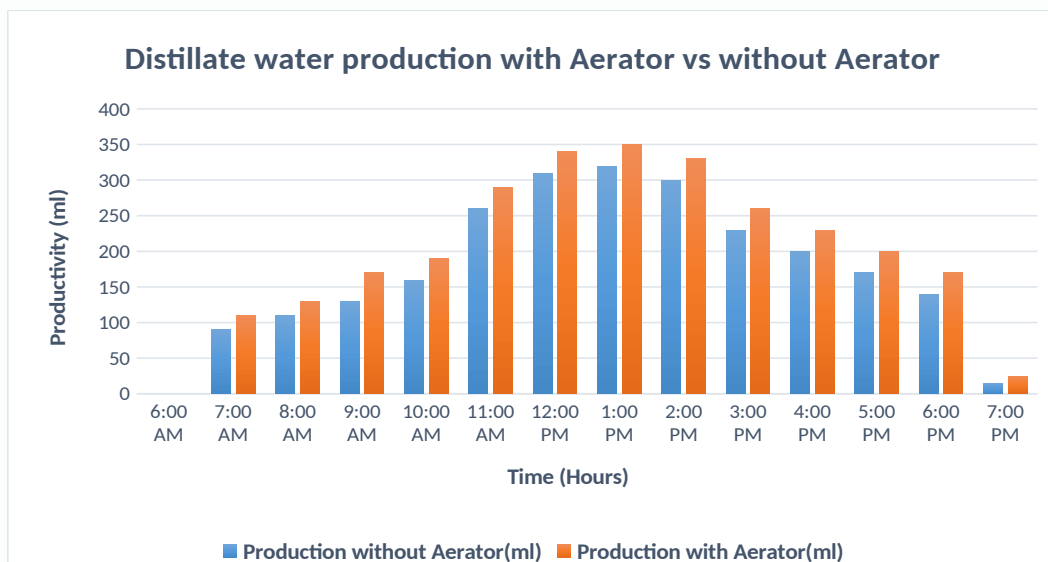


Figure 46: Shows the Hourly distillate water productivity by desalination with aerator vs without aerator of the [Date: 25-May-24].

[Date: 30-May-24].

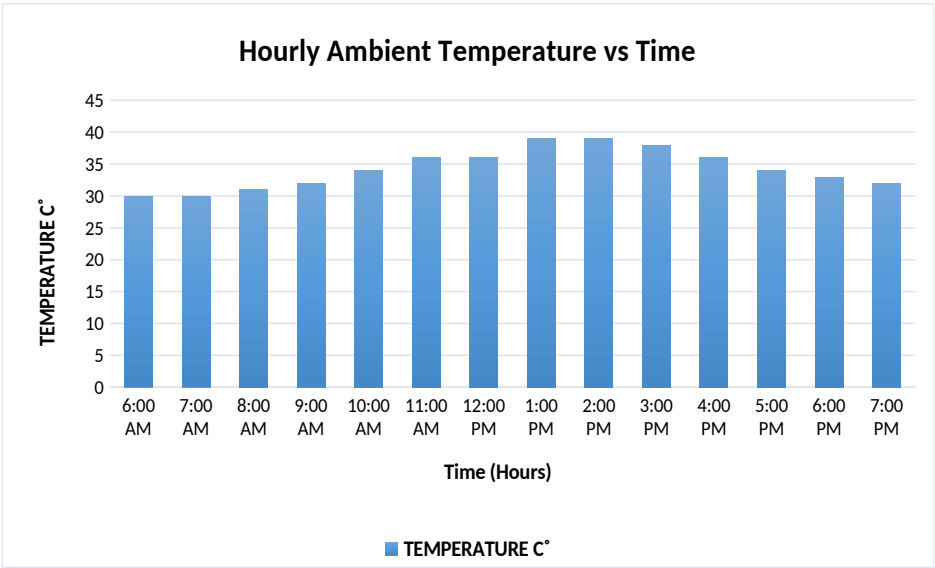


Figure 47: Shows the Hourly ambient temperature vs time of the [Date: 30-May-24].

[Date: 30-May-24].

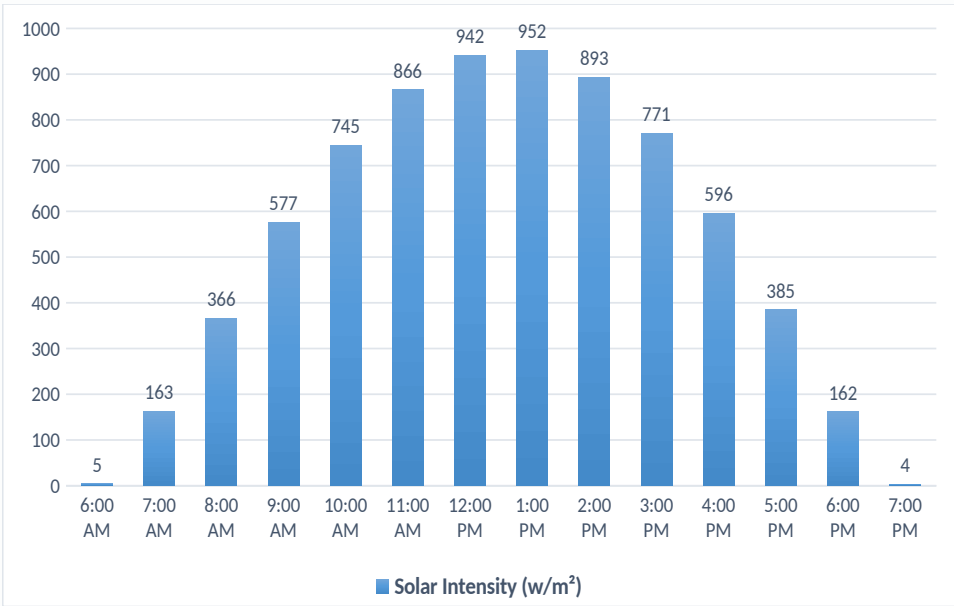


Figure 48: Shows the Hourly Solar Intensity vs Time of the [Date: 30-May-24].

[Date: 30-May-24].

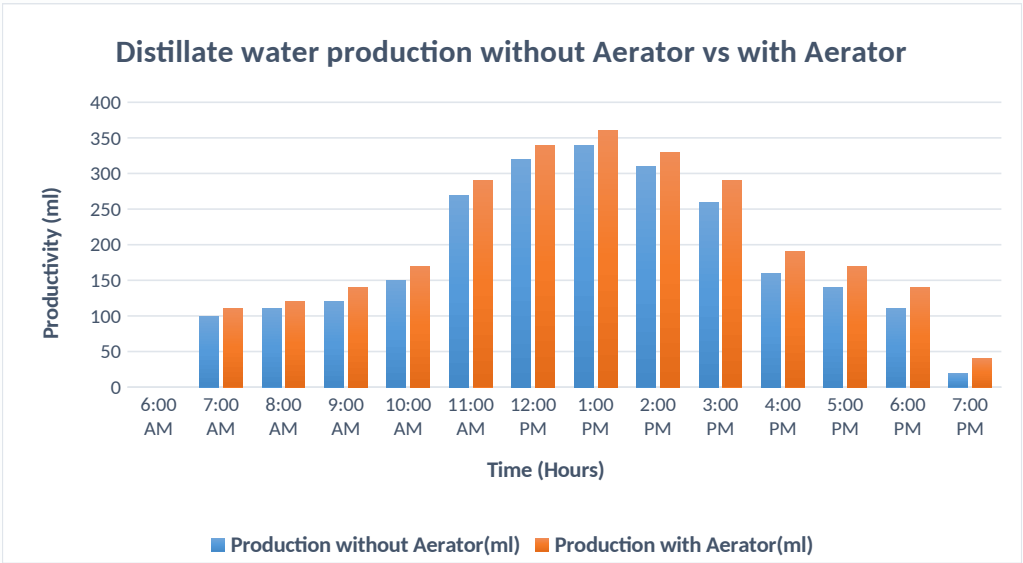


Figure 49: Shows the Hourly distillate water productivity by desalination with aerator vs without aerator of the [Date: 30-May-24].

[Date: 5-June-24].

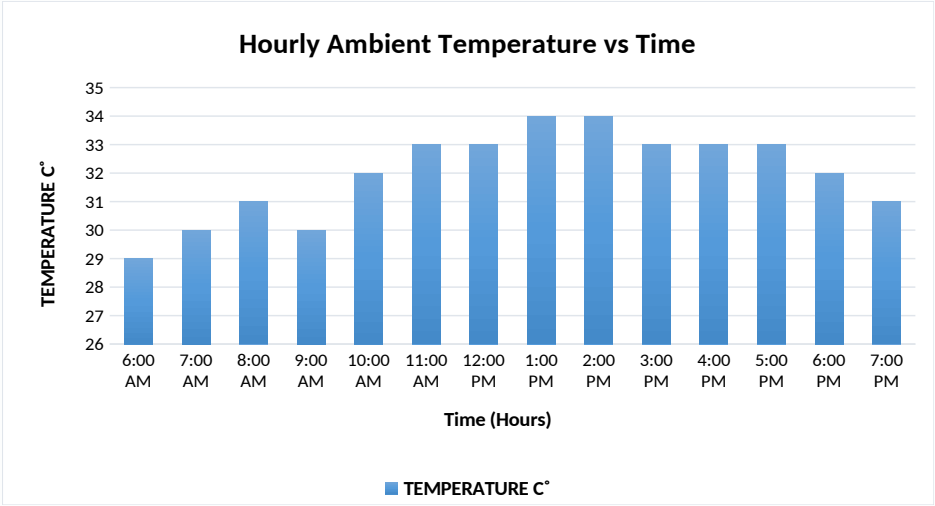


Figure 50: Shows the Hourly ambient temperature vs time of the [Date: 5-June-24].

[Date: 5-June-24].

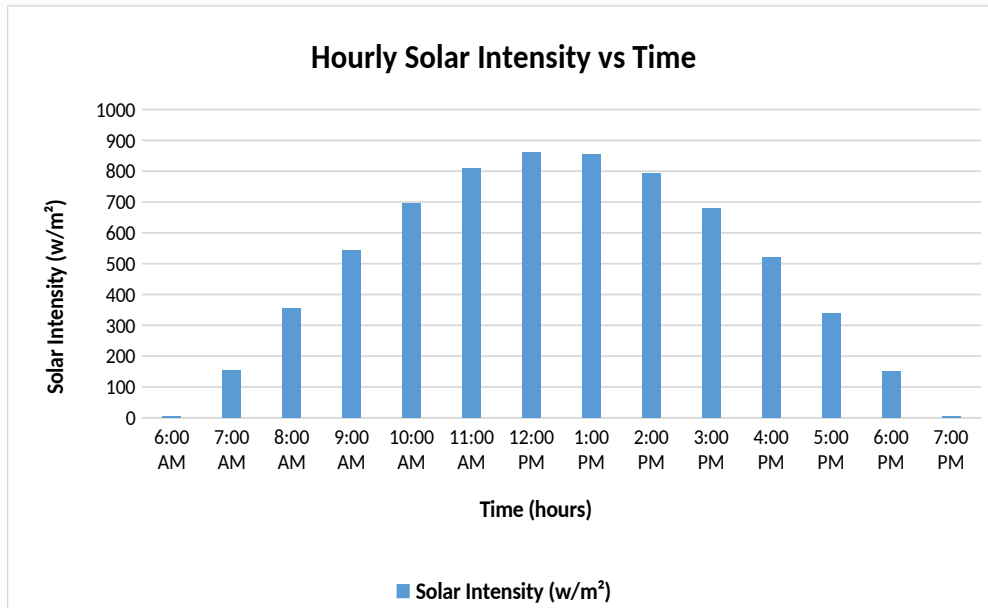


Figure 51: Shows the Hourly Solar Intensity vs Time of the [Date: 5-June-24].

[Date: 5-June-24].

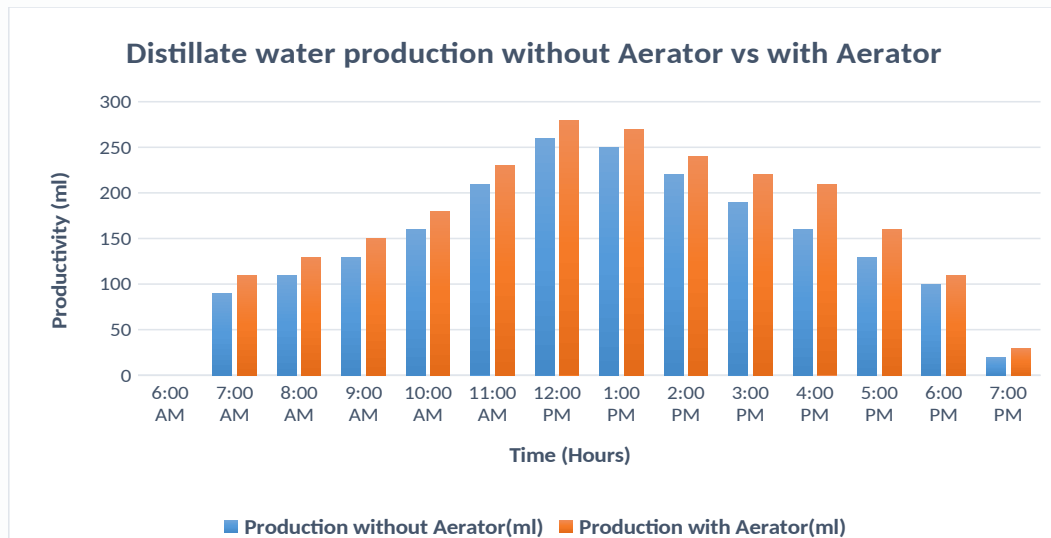


Figure 52: Shows the Hourly distillate water productivity by desalination with aerator vs without aerator of the [Date: 5-June-24].

Date: 10-June-24].

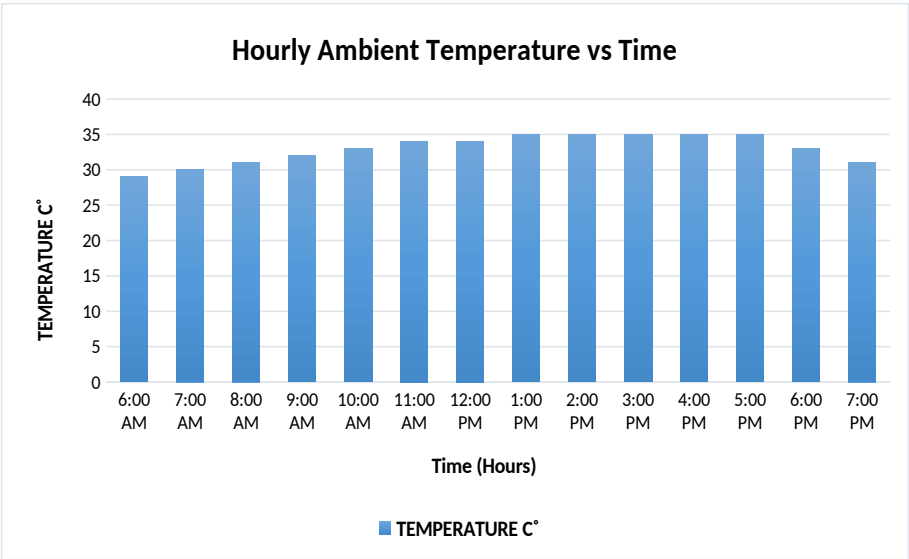


Figure 53: Shows the Hourly ambient temperature vs time of the [Date: 10-June-24].

[Date: 10-June-24].

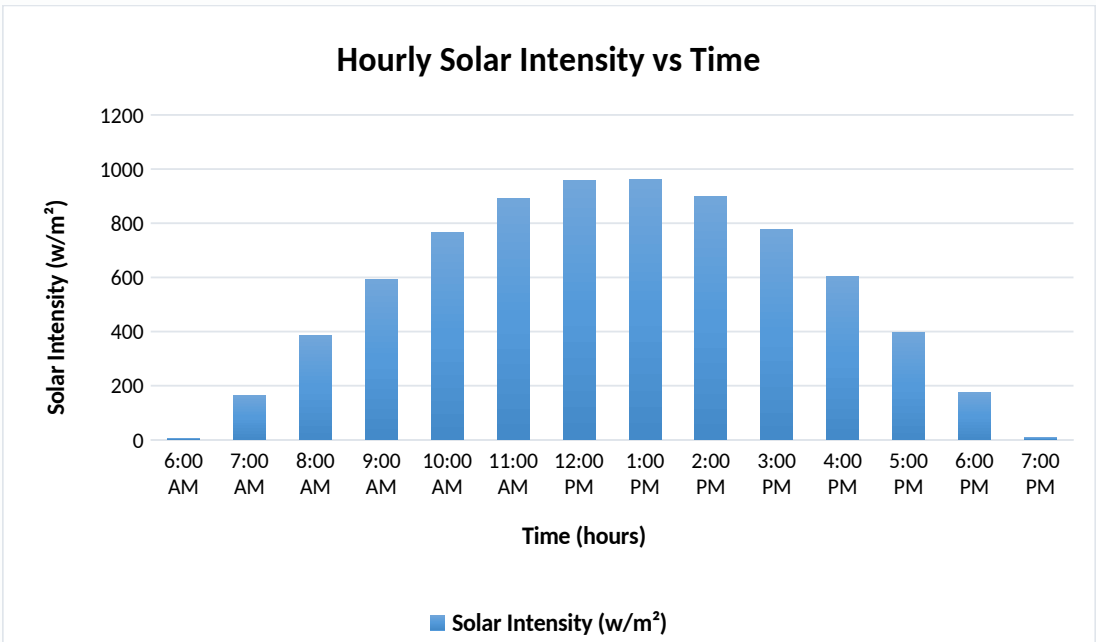


Figure 54: SHowS the Hourly Solar Intensity vs Time of the [Date: 10-June-24].

[Date: 10-June-24].

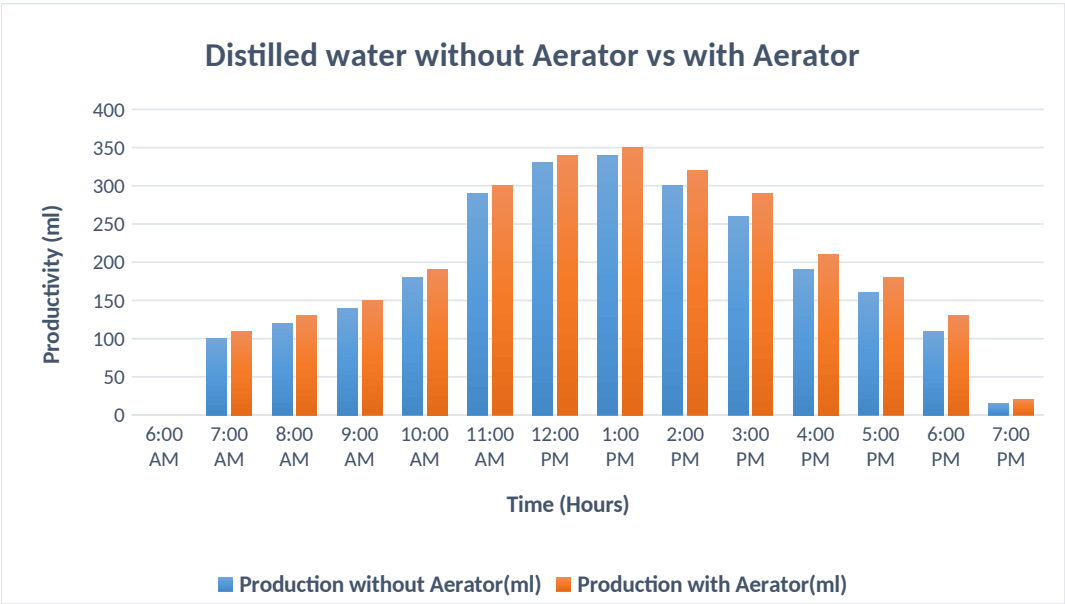


Figure 55: Shows the Hourly distillate water productivity by desalination with aerator vs without aerator of the [Date: 10-June-24].

[Date: 20-June-24].

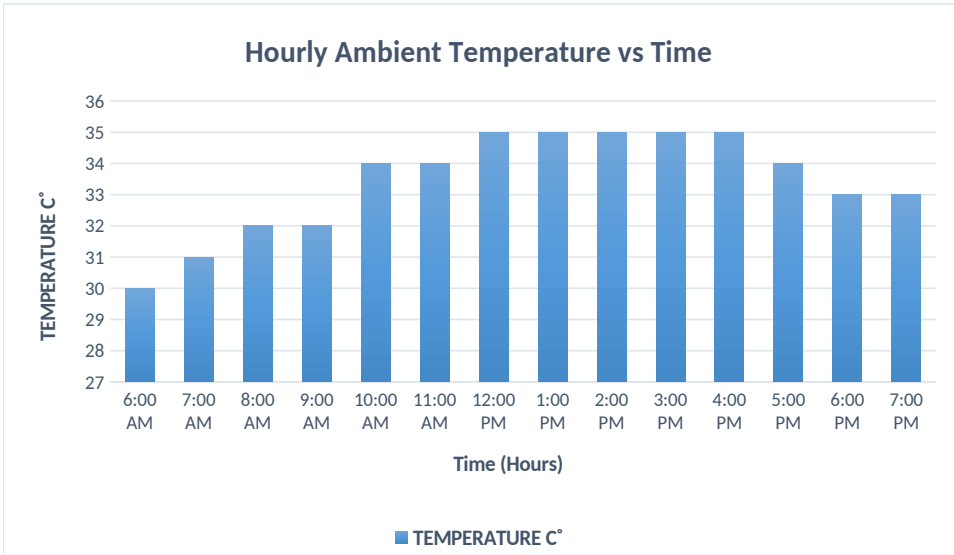


Figure 56: Shows the Hourly ambient temperature vs time of the [Date: 20-June-24].

[Date: 20-June-24].

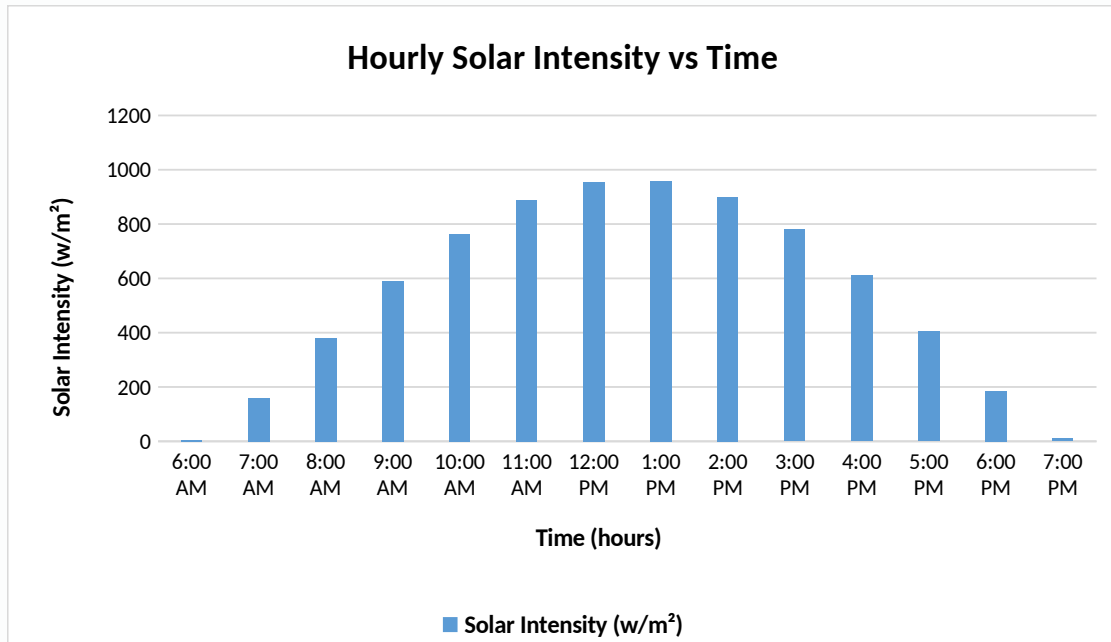


Figure 57: Shows the Hourly Solar Intensity vs Time of the [Date: 20-June-24].

[Date: 20-June-24].

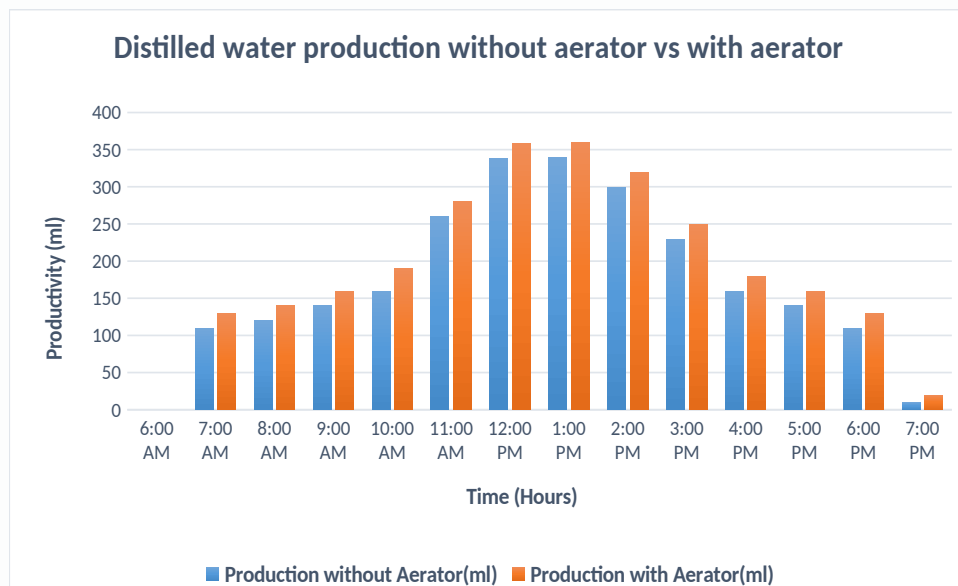


Figure 58: Shows the Hourly distillate water productivity by desalination with aerator vs without aerator of the [Date: 20-June-24].

[Date: 25-June-24].

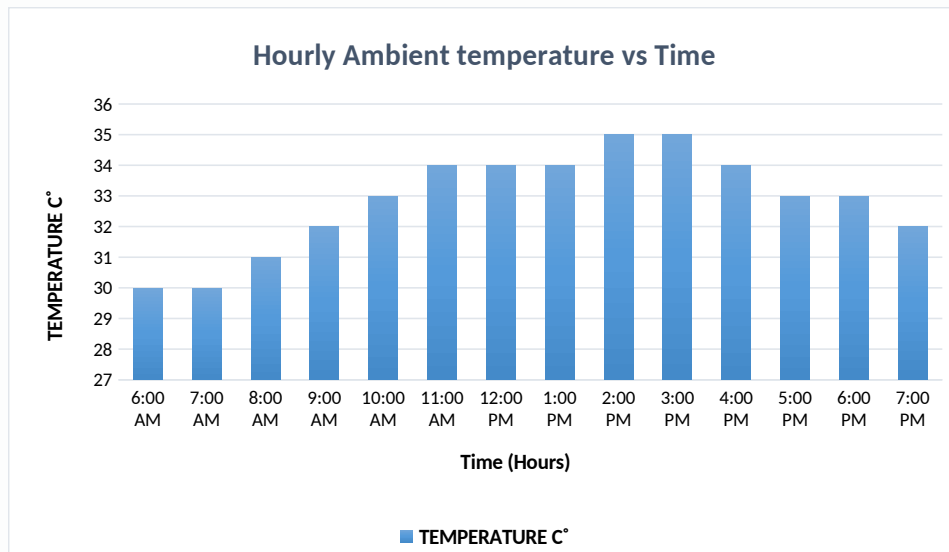


Figure 59: Shows the Hourly ambient temperature vs time of the [Date: 25-June-24].

[Date: 25-June-24].

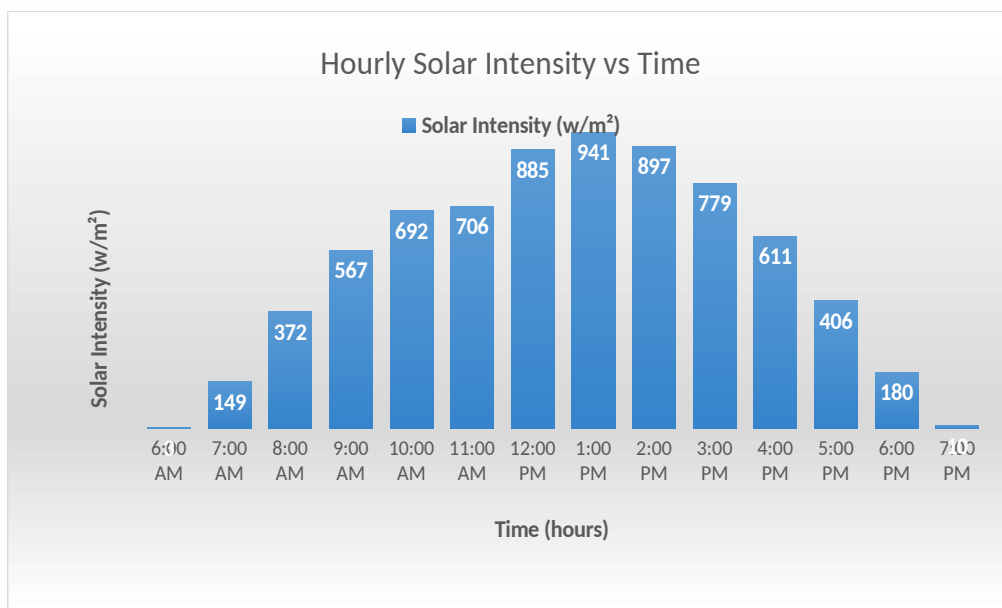


Figure 60: Shows the Hourly Solar Intensity vs Time of the [Date: 25-June-24].

[Date: 5-June-24].

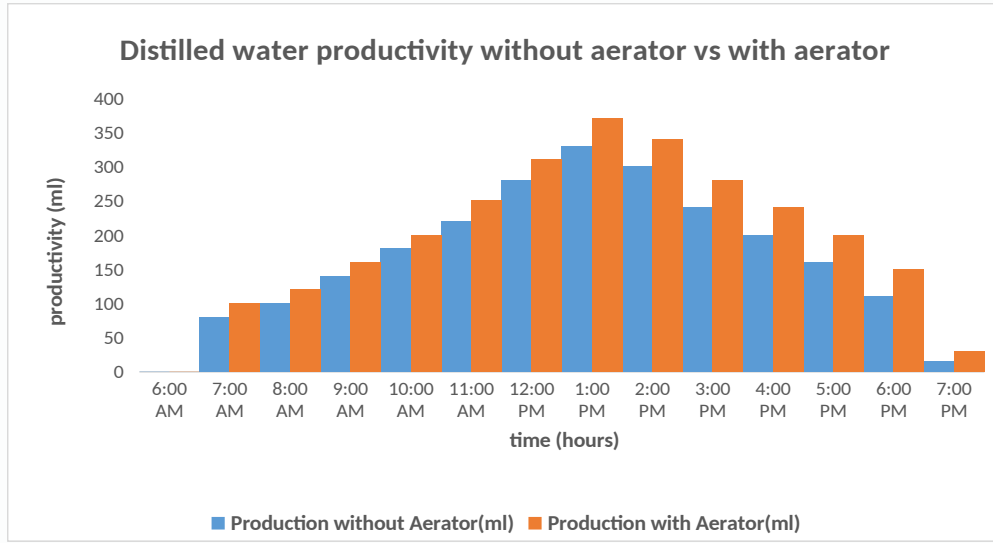


Figure 61: Shows the Hourly distillate water productivity by desalination with aerator vs without aerator of the [Date: 5-June-24].

7.2. THEORETICAL RESULTS

Here is the theoretical result for a single day is calculated taking the average temperature to investigate the difference between the results from experiments and the results from theoretical solution for the purpose of validation.

$$W = \frac{Q_{water}}{L_{vaporization}} \quad (A)$$

$$Q_{water} = Q_i - Q_{loss} \quad (B)$$

$$Q_i = A \cdot I \cdot 3 \cdot T \quad (C)$$

$$Q_{loss} = Q_{glass} + Q_{base} \quad (D)$$

$$Q_{glass} = Q_{conv, glass} + Q_{rad, glass} + Q_{cond, glass}$$

$$Q_{conv, glass} = h_{conv, glass} \cdot 3 \cdot A_{glass} \cdot 3(T_{glass} - T_{ambient})$$

$$Q_{rad, glass} = \epsilon_{glass} \cdot 3 \cdot A_{glass} \cdot 3(T_{glass}^4 - T_{ambient}^4)$$

$$Q_{cond, glass} = K_{glass} \cdot 3 \cdot A_{glass} \cdot 3\left(\frac{T_{glass} - T_{ambient}}{L_{glass}}\right)$$

: - L is the thickness of the glass

$$Q_{cond, glass} = 0.8 \frac{W}{m \cdot K} 3 \ 0.72 \ m^2 \ 3 \left(\frac{273+45-273+39}{0.004 \ m} \right) k$$

$$Q_{cond, glass} = 864 W$$

$$Q_{rad, glass} = 0.9 \ 35.67310^{-8} \frac{W}{m^2 \cdot K^4} 30.72 \ m^2 \ 3(318^4 - 312^4) K^4$$

$$Q_{rad, glass} = 0.095 \ W$$

$$Q_{conv, glass} = 15 \frac{W}{m^2 \cdot K} 3 \ 0.72 \ m^2 \ 3(318 - 312) k$$

$$Q_{conv, glass} = 108 \ W$$

$$Q_{glass} = 864 W + 0.095 \ W + 108 \ W$$

$$Q_{glass} = 972 \ W$$

$$Q_{base} = K_{base} 3 A_{base} 3 \left(\frac{T_{base} - T_{ground}}{L_{base}} \right)$$

: - L is the thickness of the base

$$Q_{base} = 2 \frac{W}{m \cdot K} 3 \ 0.72 \ m^2 \ 3 \left(\frac{321 - 303}{0.008 \ m} \right) k$$

$$Q_{base} = 324 \ W$$

$$Q_{loss} = 972 \ W + 324 W = 1296 \ W$$

$$Q_i = A 3 I 3 T$$

$$Q_i = 0.72 \ 3 \ 488 \ 3 \ 0.8 \ 3 \ 10$$

$$Q_i = 2810 \ W$$

$$Q_{water} = Q_i - Q_{loss}$$

$$Q_{water} = 2810 - 1296 = 1514 \ W$$

$$W = \frac{Q_{water}}{L_{vaporization}} = \frac{1514}{0.627} = 2414 \ ml$$

The experimental results 2.39 liters and theoretical results 2.4 liters closely matched and this strong agreement validates the theoretical model and experimental methodology used in our

Results & Discussion

Chapter 8

Interpretation of water chemistry, monthly productivity, performance drivers and economic assessment.



SOLAR-POWERED DESALINATION WITH FORCED-AIR BUBBLING

project. The close correlation between the two results shows the accuracy of our approach and provides confidence in the predicted water production rates.

CHAPTER 8.

8.1. RESULTS AND DISCUSSION

This chapter focuses on the experimental testing of the solar desalination system and details the results obtained. The analysis covers several aspects of system performance, including test results, Water chemistry before and after the experiments, Monthly Productivity of water and economic analysis.

8.1.1. Test results:

The test results obtained in the test phase are presented and analyzed to investigate the functionality of the solar desalination system.

8.1.2. Water chemistry before experiment:

The sea water properties like pH, conductivity and turbidity are noted before the experiment which are shown below in the table 2.

Table 2: Shows the chemistry of sea water

Parameters	Values
pH	7.5 – 8.4
TDS	30,000 – 38,000 mg/L
Conductivity	45,0000 – 55,000 μ S/cm

8.1.3. Chemistry of Distilled Water after experiment:

The distilled water properties like pH, conductivity and turbidity are noted after the experiment which are shown in the table 3 below.

Table 3: Shows the chemistry of distilled water

Parameters	Values
pH	6.5 – 8.0
TDS	6 mg/L
Conductivity	12 μ S/cm

8.1.4. Monthly Water Production:

The monthly water productivity is meticulously observed and noted on the basis of daily experiment and then it is found the average distillate water production for the period of four moths each as shown in tables 4, 5, 6 and 7 below.

I. Monthly Water Production For March:

Table 4: Shows the average production of distillate water for the month of March.

Date	Solar Intensity (W/m ²)	Production without Aerator (ml)	Production with Aerator (ml)	Efficiency
24-Mar-24	6428	1940	2300	18.55670103
25-Mar-24	6657	2110	2365	12.08530806
26-Mar-24	6671	2050	2315	12.92682927
27-Mar-24	6383	2195	2365	7.744874715
28-Mar-24	6691	2095	2385	13.8424821
29-Mar-24	6682	2010	2290	13.93034826
30-Mar-24	6701	2205	2415	9.523809524
31-Mar-24	6710	2085	2310	10.79136691

Total		16690	18745	12.31276213
Average		2086.25	2343.12	12.3126213

II. Monthly Water Production For April:

Table 1: Shows the average production of distillate water for the month of April.

Date	Solar Intensity (W/m ²)	Production without Aerator (ml)	Production with Aerator (ml)	Efficiency
1-Apr-24	7099	2134	2315	8.48172446
2-Apr-24	7116	2144	2405	12.1735075
3-Apr-24	7109	2140	2470	15.4205608
18-Apr-24	7118	2125	2360	11.0588235
19-Apr-24	7121	2245	2450	9.13140312
20-Apr-24	7140	2025	2270	12.0987654
21-Apr-24	7155	2250	2530	12.4444444
22-Apr-24	7122	2140	2435	13.7850467
24-Apr-24	7167	2190	2465	12.5570776
25-Apr-24	7272	2240	2505	11.8303571
Total		21633	24205	11.8892433
Average		2163.3	2420.5	11.8892433

III. Monthly Water Production For May:

Table 6: Shows the average production of distillate water for the month of May

Date	Solar Intensity (W/m ²)	Production without Aerator (ml)	Production with Aerator (ml)	Efficiency
1-May-24	7377	2485	2755	10.86519115
2-May-24	6606	2215	2590	16.93002257
3-May-24	3625	1845	2070	12.19512195
4-May-24	7197	2485	2745	10.46277666
5-May-24	7403	2375	2740	15.36842105
6-May-24	7426	2490	2780	11.64658635
7-May-24	7436	2495	2790	11.82364729
8-May-24	7444	2630	2850	8.365019011
9-May-24	7452	2660	2890	8.646616541
10-May-24	7457	2550	2790	9.411764706
11-May-24	7466	2530	2760	9.090909091

12-May-24	7298	2445	2680	9.611451943
13-May-24	7464	2525	2750	8.910891089
14-May-24	7052	2360	2690	13.98305085
15-May-24	7216	2460	2710	10.16260163
20-May-24	7528	2489	2760	10.88790679
21-May-24	7532	2470	2770	12.14574899
25-May-24	7555	2435	2795	14.78439425
26-May-24	7498	2440	2790	14.3442623
27-May-24	7540	2345	2715	15.7782516
28-May-24	7571	2405	2700	12.26611227
29-May-24	7531	2375	2635	10.94736842
30-May-24	7427	2410	2690	11.61825726
31-May-24	7562	2440	2780	13.93442623
Total		65939	73585	11.5955656
Average		2442.185185	2725.37037	11.5955656

IV. Monthly Water Production For June:

Table 7: Shows the average production of distillate water for the month of June.

Date	Solar Intensity (W/m ²)	Production without Aerator (ml)	Production with Aerator (ml)	Efficiency
1-Jun-24	7577	2395	2750	14.82255
2-Jun-24	7553	2415	2760	14.28571
3-Jun-24	7582	2415	2750	13.87164
4-Jun-24	7566	2465	2780	12.7789
5-Jun-24	6770	2030	2320	14.28571
6-Jun-24	6674	1995	2295	15.03759
7-Jun-24	6943	2365	2610	10.35941
8-Jun-24	7600	2410	2660	10.37344
9-Jun-24	7600	2465	2780	12.7789
10-Jun-24	7589	2535	2720	7.29783
11-Jun-24	7604	2475	2730	10.30303
12-Jun-24	7489	2375	2700	13.68421
13-Jun-24	7573	2455	2690	9.572301
14-Jun-24	7608	2419	2679	10.74824
15-Jun-24	7166	2460	2710	10.1626
20-Jun-24	7587	2419	2679	10.74824
21-Jun-24	6205	2300	2690	16.95652
22-Jun-24	7315	2465	2710	9.939148
23-Jun-24	7563	2405	2675	11.22661
24-Jun-24	7565	2535	2745	8.284024
25-Jun-24	7198	2355	2750	16.77282
Total		50153	56183	12.02321
Average		2388.238095	2675.380952	12.02321

8.1.5. Total Productivity:

The total productivity of a solar desalination system, measured in freshwater production volume, is calculated and compared under different operating conditions. The maximum water production is noted in the month of May which is due to the ambient temperature. The average water production by solar still with aerator is noted is 2.7L/day, while without aerator is 2.4L/day. It is observed that increase in temperature and solar intensity increases the water productivity.

8.1.6. Monthly Production Averages

The average daily production of freshwater using the solar desalination system varied across the four months as in table 8 follows:

Table 8 shows the monthly water production with aerator and without aerator

Monthly water production with aerator and without aerator	March	April	May	June
Production with aerator (L)	2.3	2.4	2.7	2.6
Production without aerator(L)	2	2.1	2.4	2.3

8.1.7. Factors Affecting the Production of water:

The performance of the solar powered desalination system is disturbed by some variables, like radiation from sun, temperature, humidity, speed of wind, water depth, and the presence of the aerator. Each factor's impact is detailed as follow:

I. Solar Radiation:

Radiation from the sun is the main source of energy driving the desalination system. Higher solar radiation increases the heat absorbed by the water, which is leading to higher evaporation rates.

i. March:

As the transition month from winter to spring, solar radiation levels begin to increase.

ii. April:

Higher solar radiation due to longer daylight hours.

iii. May:

Peak solar radiation, significantly boosting productivity.

iv. June:

High solar radiation, though slightly less than May due to occasional cloudy weather.

II. Ambient Temperature:

Higher ambient temperatures enhance the evaporation rate by providing additional heat to the water in the basin

i. March:

Moderate temperatures as the weather warms up.

ii. April:

Increasing temperatures contribute to higher evaporation rates.

iii. May:

High temperatures, coupled with peak solar radiation, maximize productivity.

iv. June:

High temperatures, though humidity starts to increase, slightly affecting productivity.

III. Humidity:

Humidity affects the evaporation rate; higher humidity levels reduce the evaporation potential.

i. March:

Lower humidity favors higher evaporation rates.

ii. April:

Still relatively low humidity, maintaining high productivity

iii. May:

Moderate humidity, with a slight reduction in evaporation efficiency.

iv. June:

Higher humidity reduces evaporation rates, slightly decreasing productivity.

IV. Wind Speed:

Wind speed aids in the removal of the water vapor from the basin, enhancing the condensation process.

i. March:

Moderate wind speeds help maintain steady productivity.

ii. April:

Increased wind speeds in spring improve vapor removal.

iii. May:

Optimal wind conditions further boost productivity.

iv. June:

Variable wind speeds can either enhance or slightly reduce productivity.

V. Water Depth:

The optimal water depth ensures maximum heat absorption and efficient evaporation. Too much water can decrease the temperature rise, while too little water reduces the thermal mass. The study found 2 cm to be the optimal depth, and this was maintained consistently across all months.

VI. Aerator Effect:

The integration of an aerator significantly improved productivity by enhancing the water-air interface, increasing the evaporation rate.

March to June consistently, the aerated system outperformed the non-aerated system by approximately 0.3 to 0.4 liters daily, demonstrating the effectiveness of aeration in boosting freshwater production.

The results demonstrate a clear seasonal variation in the production of the solar desalination system, heavily influenced by climatic factors. The highest productivity was observed in May, attributed to peak solar radiation and high ambient temperatures. June showed a slight decrease in productivity compared to May, likely due to increased humidity and occasional cloudy weather.

The consistent higher productivity of the system with an aerator across all months highlights the significant advantage of integrating an aerator into the design which was our aim. The aerator enhances the evaporation process, thereby increasing the overall freshwater yield. This advantage is particularly noticeable during the months with slightly lower solar radiation and higher humidity, where the aerator helps maintain higher productivity levels.

In conclusion, the study underscored the importance of optimizing design parameters, such as water depth and the use of an aerator, to enhance the performance and productivity of solar stills. The findings provide valuable insights for the designing of more effective and less cost solar desalination technologies tailored to specific climatic conditions.

8.1.8. Economical analysis:

Table 2. Above present the expenses of solar powered desalination systems as we designed two systems one with the integration of aerator and the second one is without aerator.

The total cost of both systems is PKR 52,000. These two units of solar still were fabricated for the purpose of research.

Now, we will do the economic analysis of the system. The aim of this study is to find out the cost of 1 liter of distilled water from the system based on the experiments. Furthermore, a comparison would be taken on cost of one liter of distilled water obtained from solar still without aerator and solar still with aerator.

The newly designed solar still is supposed to have a life of 10 years. The cost that would be included during these years can be grouped into fixed cost and variable cost. The fixed cost is the cost of designing of solar still that is to be accounted only at once. The variable cost includes the maintenance cost to keep the solar still in the running condition. The variable cost that is to be accounted per year is estimated by 5% of the fixed cost.

Assumed that the system will be operated for 290 days a year. The remaining days are accounted in weekly maintenance (52 days), cloudy and rainy days. Maintenance would consist of the removing of salts from the tank, cleaning, washing of glass cover, changing of any damaged parts, and repairing of some parts if needed.

The cost of one liter of distilled water is calculated by summing all the costs (fixed and variable) and divided by total production by solar still in 10 years. From the results of experiments, the production of solar still per day with aerator and without aerator is found to be 2.8 L/day and 2.4 L/day, respectively. It is supposed that production will be the same throughout the 10 years.

Table below presents the calculation of cost of 1 liter of distilled water obtained from solar desalination system with and without aerator. Cost of 1 liter of freshwater from solar still with and without aerator is found to be 4.98 Rs/L and 5.38 Rs/L respectively. The results shows that the solar system with aerator is more economically suitable than solar system without aerator as shown in the table 9 below.

Table 9: Shows the Cost of 1 liter of fresh water obtained from solar still with aerator and without aerator.

SOLAR POWERED DESALINATION SYSTEM WITH SINGLE SLOPE SOLAR STILL	WITH AERATOR	WITHOUT AERATOR
Life cycle (years)	10	10
Fixed cost (Rs)	27000	25000
Variable cost (Rs)	13500	12500
Total cost (Rs)	40500	37500
Average Daily Productivity (L/day)	2.8	2.4
Annual Productivity (L/year)	812	696
Lifetime Productivity (L)	8120	6960
Cost Per Liter (Rs/L)	4.98	5.38

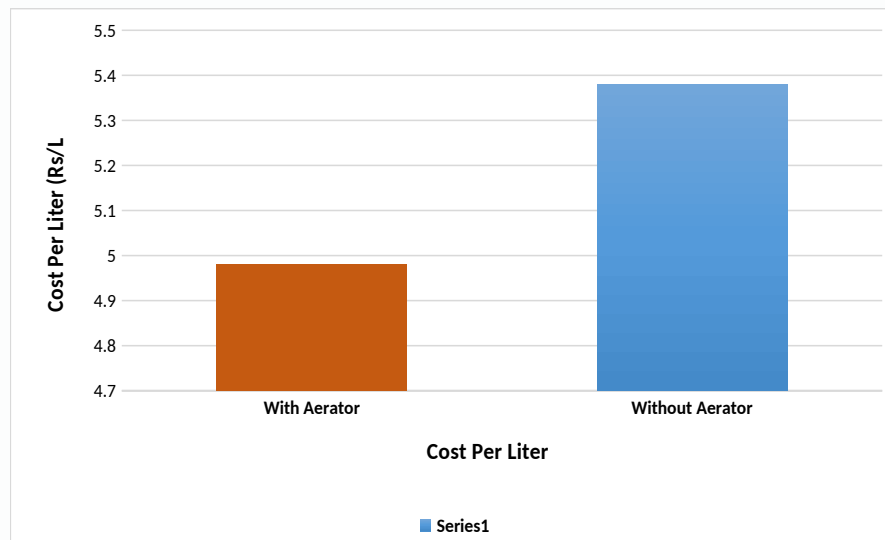


Figure 62:Shows the Comparison of cost of 1 liter of distilled water obtained from solar still with aerator and without aerator.

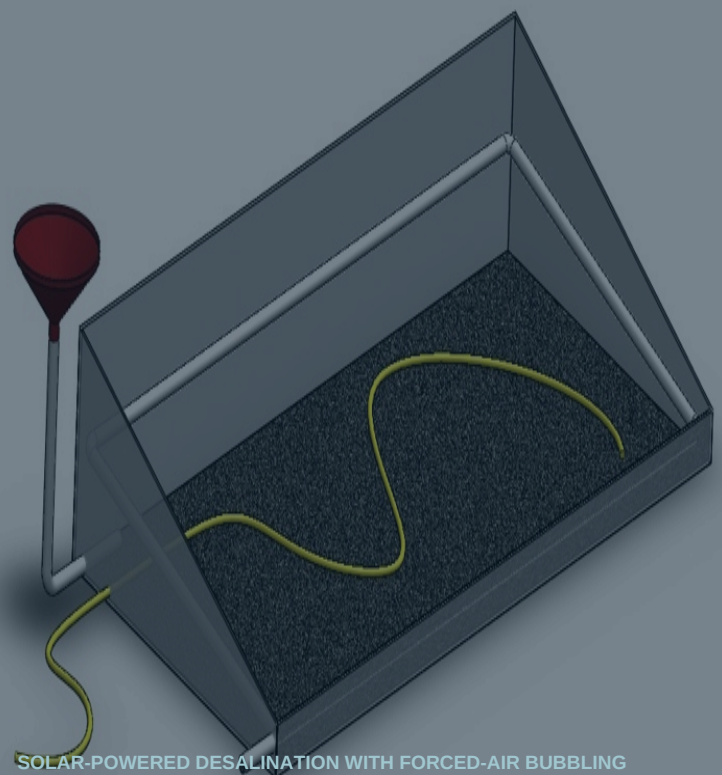
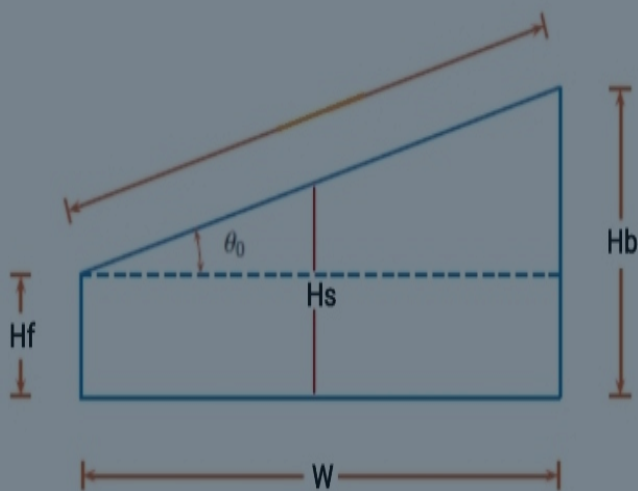
SECTION

Integrated Findings

Conclusion

Consolidated findings from the design, modelling, experimental comparison and economic evaluation.

(b)



CONCLUSION

The project aimed to address the critical issue of clean water scarcity by developing a desalination system run by solar energy integrated with an air-forced bubble effect through an aerator. This system, designed to suit the climatic conditions of Karachi, Pakistan, demonstrated significant potential for broader application across the Asian continent. The primary objectives included the design, construction, and evaluation of the solar desalination system.

A mathematical model was developed using mathematical equations for the various parts of the solar still. The graphs for the basin, outer and inner glass surfaces, and water obtained from the model closely matched the experimental data, confirming the model's accuracy. The system's design was meticulously crafted using DS SolidWorks, incorporating initial design parameters and material specifications. Following comprehensive market research, the solar panel was fabricated, and a series of experimental studies were conducted. Comparative analysis of systems with and without aeration showed that the aerated system was more productive and cost-effective, yielding 2.8 liters of fresh water per day compared to 2.4 liters for the non-aerated version.

Economic feasibility was assessed, revealing that the cost of 1 liter of distilled water from the aerated system was 4.98 PKR, slightly lower than the 5.38 PKR per liter for the non-aerated system. The analysis included fixed costs (initial fabrication) and variable costs (annual maintenance), with an operational expectation of 290 days per year over a lifespan of 10 years.

The project's findings underscore the enhanced productivity and economic feasibility of the aerated solar still. This system not only provides a viable solution for clean water production but also aligns with sustainable and cost-effective practices. The significant reduction in total dissolved solids (TDS) from 35000 to 150 mg/L highlights the system's efficiency. In summary, the developed solar desalination system with an aerator demonstrates considerable promise as an easy solution for solving water scarcity problems, offering improved performance and economic benefits over traditional solar stills. This innovative approach, validated through both theoretical modeling and practical experimentation, provides a scalable model for enhancing water availability in arid and semi-arid regions.

SECTION

Research Outlook

Future Prospects

Scaling, technological improvement, renewable integration, community deployment and opportunities for future research.



SOLAR-POWERED DESALINATION WITH FORCED-AIR BUBBLING

FUTURE PROSPECTS

This project explores the future outlook for solar desalination systems and potential opportunities for further development and implementation. It explores various dimensions including scalability, technological improvement, integration with renewable energy sources, community engagement, political support and research innovation to map industrial advancement and effective solutions to global water challenge

Scaling and Deployment

Options for scaling up the solar desalination system for large-scale deployment are being explored. Strategies to help overcome scalability issues such as cost effectiveness, infrastructure requirements, and environmental considerations are discussed. The chapter examines the technology's feasibility in various geographic areas and contexts, including remote communities, arid regions, and disaster zones.

Technological Improvement:

Technological innovations will explore ways to enhance the functionality, efficiency and reliability of a solar powered desalination system. Areas of focus may include research into advanced materials, new system designs, innovative heat transfer mechanisms, and integrated system optimization. The chapter discusses the potential impact of technological development on the affordability, sustainability and accessibility of the system.

Integration with renewable energy sources:

The future prospect explores the possible synergies and benefits of integrating a solar desalination system with other energy sources such as wind energy, biomass energy and thermal power. Strategies to hybridize desalination technologies to maximize energy efficiency, reliability, and sustainability are being explored. The chapter emphasizes the importance of energy-water integrated planning and interdisciplinary cooperation in exploiting the full potential of renewable energy-based desalination solutions.

Community Engagement and Capacity Building:

The importance of community involvement, capacity building and knowledge sharing is emphasized in promoting the adoption and acceptance of solar desalination technologies. Strategies to raise awareness, promote local ownership, and develop technical skills and knowledge in communities are discussed. The chapter explores the role of education and participatory approaches in enabling communities to sustainably meet their water needs.

Policy and Regulatory Support:

The policy and regulation discuss the need for supportive policy frameworks, regulations and incentives to facilitate the widespread adoption and deployment of solar desalination systems.

Policy recommendations are made to promote investment, innovation and market development in the desalination sector. The chapter examines the role of government agencies, regulatory bodies and international organizations in promoting an enabling environment for sustainable water supply and technology adoption.

Research and Innovation:

It emphasizes the importance of continuous research, innovation and collaboration in advancing solar desalination technology. Key research areas such as membrane development, energy storage, brine management and system optimization are defined. The chapter calls for increased funding, networking and information exchange to support interdisciplinary research projects designed to respond to pressing water challenges and achieve water security. By exploring these perspectives and strategies, this chapter aims to inspire action, collaboration and innovation to harness the prospect of solar powered desalination system as a durable solution to global scarcity of water and ensure access to clean water for all.

APPENDIX A

TIME SCALE:

Table 10: Shows the specific time for project.

No.	Elapsed time from start (in months) of the project	Milestone
1	August to September	Project Initiation
2	October	Literature Review and Research
3	November	Technology Selection and Design
4	December	Procurement and Preparation
5	January to February	Assembly, Installation and Optimization
6	March to June	Testing and Analysis
7	July	Documentation and Reporting

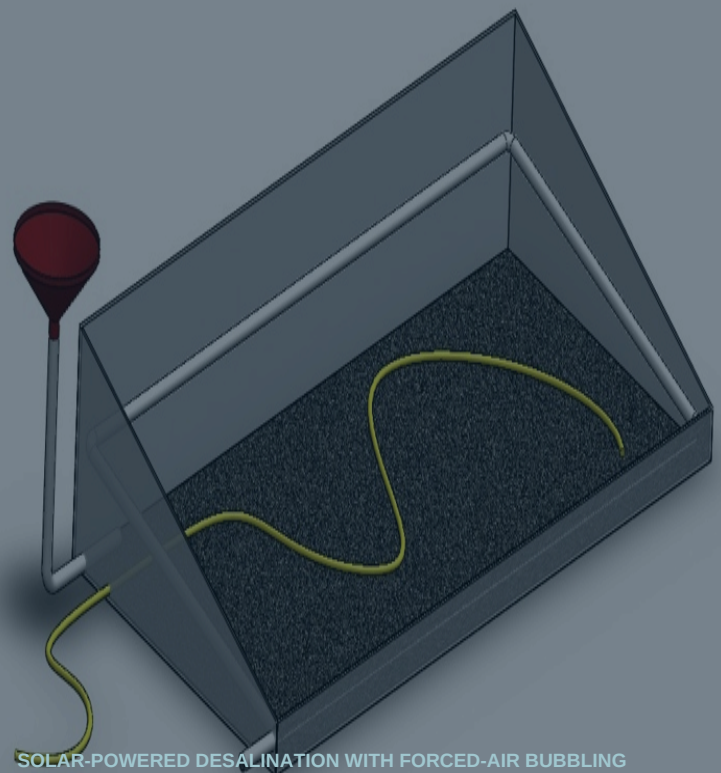
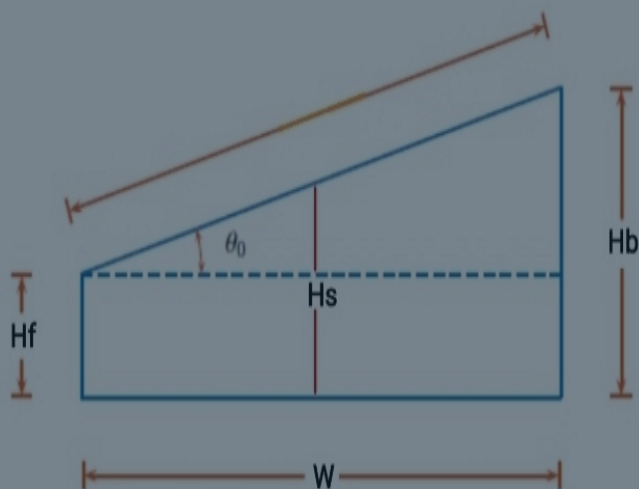
SECTION

Project Records and Sources

Appendix & References

Project timing records, Gantt chart and the original reference list.

(b)



GANTT CHART:

Table 11: Shows the monthly working chart

Milestone	AUG.	SEP.	OCT.	NOV.	DEC.	JAN.	FEB	MAR.	APR.	MAY	JUN.	JUL.
Project Initiation												
Literature Review and Research												
Technology Selection and Design												
Procurement and Preparation												
Assembly, Installation and Optimization												
Testing and Analysis												
Documentation and Reporting												

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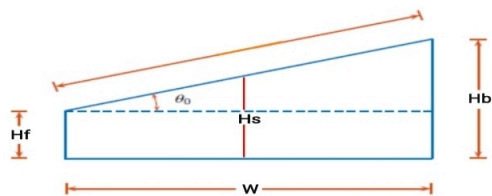
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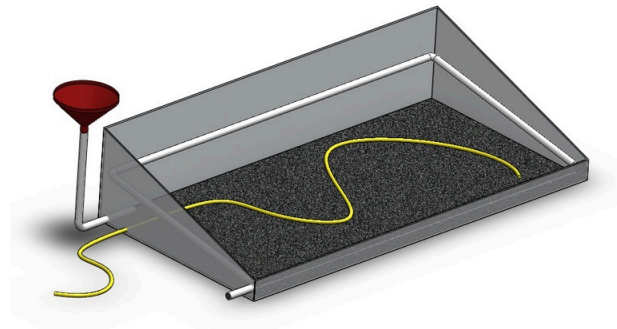
About the Author

Abdul Murad is a Mechanical Engineering graduate whose undergraduate research focused on solar-powered desalination with forced-air bubbling. The work documented in this monograph combines CAD-based design, fabrication, mathematical modelling, experimental testing, water-quality assessment and economic evaluation. This academic edition was prepared to present the complete research record in a clear, professional format for scholarly discussion, graduate-study applications and research portfolios.

(a)



(b)



Research domain

Solar thermal desalination | experimental energy systems | freshwater productivity | sustainable engineering



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